This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

DHPCL: Diffusion-Driven Hard-Soft Prototype Contrastive Learning for Semi-Supervised Medical Image Segmentation

Xiaogang Du, *Member, IEEE*, Chunliang Li, Tongfei Liu, *Member, IEEE*, Tao Lei, *Senior Member, IEEE*, Yingbo Wang, and Asoke K. Nandi, *Fellow, IEEE*

***Abstract*— Semi-supervised learning (SSL) has emerged as a powerful paradigm for medical image segmentation, effectively mitigating the reliance on large-scale pixel-level annotations. However, existing SSL approaches frequently suffer from a vicious cycle of insufficient pseudo-label quality and degraded feature discriminability, where noisy predictions induce poor feature representations and vice versa. To address this problem, we propose a novel framework, Diffusion-Driven Hard–Soft Prototype Contrastive Learning (DHPCL) for semi-supervised medical image segmentation. First, we design a Discriminator-Guided Diffusion Rectification (DGDR) module. Unlike existing methods that identify noise via static thresholds, DGDR leverages pixel-wise confidence maps from a discriminator to guide a latent diffusion model, adaptively performing stronger reconstruction on low-confidence regions while preserving reliable structures through a Local Detail Enhancement (LDE) mechanism. Second, to address feature degradation, we propose a Hard-Soft Harmonized Prototype Contrast (HSH-PC) module. This component constructs dynamic class prototypes from diffusion-refined labels and enforces feature alignment using a dual-constraint strategy: imposing hard constraints on high-confidence pixels and soft distribution-based guidance for ambiguous regions. This harmonized approach significantly improves intra-class compactness and inter-class separability. Extensive experiments on the ACDC, MS-CMRSeg, and LA datasets demonstrate that the proposed framework achieves superior segmentation accuracy and boundary precision compared to state-of-the-art semi-supervised methods. The source code will be released after the publication of the paper.**

***Index Terms*— Semi-Supervised Learning, Medical Image Segmentation, Pseudo-Label Rectification, Diffusion Models, Prototype Contrastive Learning**

This work was supported by the Young Science and Technology Innovation Leading Talents Program of Xi'an, China (No. 25ZQRC00019), National Natural Science Foundation of China (Nos. 61861024, 62271296, and 62201334), Scientiffc Research Program Funded by Education Department of Shaanxi Provincial Government (Nos. 23JP022, and 23JP014). (Corresponding author: Tao Lei.)

Xiaogang Du, Chunliang Li, Tongfei Liu, Tao Lei, Yingbo Wang are with the Shaanxi Joint Laboratory of Artificial Intelligence and the School of Electronic Information and Artificial Intelligence, Shaanxi University of Science and Technology, Xi'an 710021, China (e-mail: duxiaogang@sust.edu.cn; 241611013@sust.edu.cn; liutongfei@sust.edu.cn; leitao@sust.edu.cn; wangyingbo@sust.edu.cn).

Asoke K. Nandi is with the Department of Electronic and Electrical Engineering, Brunel University of London, Uxbridge UB8 3PH, U.K., and also with the School of Mechanical Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: asoke.nandi@brunel.ac.uk).

I. Introduction

MEDICAL image segmentation is a fundamental process in computerized medical analysis, aiming to partition images into meaningful anatomical regions or specific regions of interest (ROIs). As a critical bridge between raw imaging data and clinical interpretation, accurate organ segmentation is a prerequisite for effective clinical decision making. By providing reliable quantitative metrics and anatomical structural information, this precise segmentation facilitates critical downstream tasks, such as early disease detection, lesion evaluation, and surgical planning [1]–[3].

With the advancement of deep learning, fully supervised segmentation methods based on CNNs [4]–[6] and Transformers have achieved remarkable performance on a variety of medical imaging tasks [7]–[9]. These methods rely on large-scale, high-quality annotated data, enabling models to learn complex structural representations. However, obtaining pixel-level annotations is labor-intensive and highly dependent on expert knowledge, particularly for tasks that involve intricate anatomical structures, where the difficulty of annotation increases exponentially [10], [11]. This limitation substantially hinders the widespread adoption of fully supervised methods and has motivated the exploration of semi-supervised learning for medical image segmentation.

Semi-supervised learning mitigates the reliance on costly annotations by jointly leveraging a small set of labeled data and a large amount of unlabeled data. Representative strategies for semi-supervised medical image segmentation mainly include pseudo-label learning [12]–[15] and consistency regularization [16]–[18]. Pseudo-label learning follows a self-training paradigm, where predictions on unlabeled images are used as supervisory signals to iteratively optimize the segmentation model. In contrast, consistency regularization encourages invariant predictions under different perturbations, augmentations, or network branches. Although these strategies have achieved considerable progress, their effectiveness remains highly dependent on prediction reliability in low-label regimes. As illustrated in Fig. 1(a), when inaccurate predictions are used as pseudo-supervision or consistency targets, they may disturb the semantic organization of the feature space, reducing intra-class compactness and blurring inter-class boundaries. The degraded feature space further

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

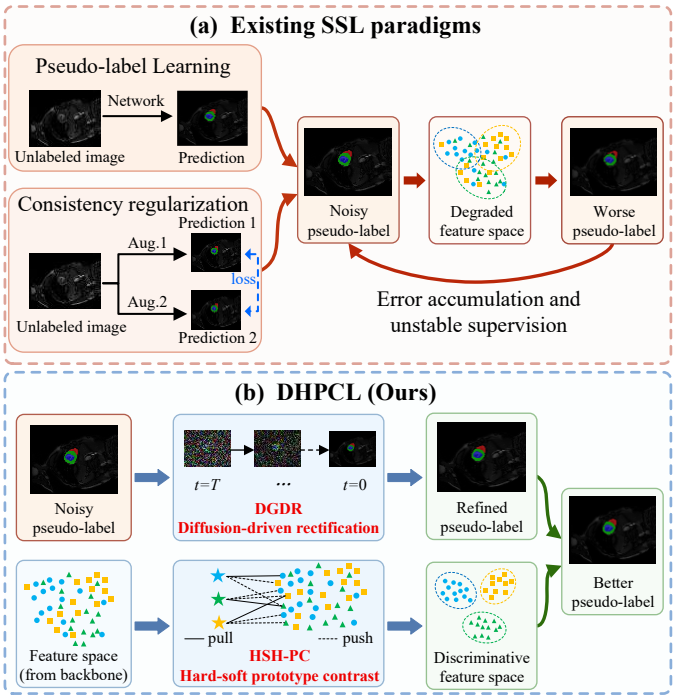


Fig. 1. Motivation of the proposed DHPCL framework. (a) In conventional semi-supervised segmentation, noisy pseudo-labels or biased consistency targets derived from unreliable predictions disturb the feature space, and the degraded feature representations further produce worse predictions, forming a self-reinforcing error accumulation process. (b) The proposed DHPCL breaks this cycle by introducing DGDR to actively restore reliable pseudo-label structures and HSH-PC to reorganize the feature space, thereby jointly improving pseudo-label quality and feature discriminability.

produces less reliable predictions, forming a self-reinforcing cycle.

Breaking this cycle remains challenging for existing methods. Confidence- or uncertainty-based pseudo-label filtering methods [12], [19] can suppress part of unreliable supervision, but they mainly discard uncertain pixels rather than actively correcting structural errors. As a result, ambiguous yet informative regions, such as weak boundaries and small anatomical structures, may be underutilized. Dynamic pseudo-label refinement methods [20]–[22] improve pseudo-label quality through staged correction, model collaboration, or boundary-aware aggregation, but they still depend on the current prediction distribution. Consistency regularization methods [23]–[28] improve robustness by enforcing agreement under perturbations or among multiple networks. However, when the consistency target is biased in low-confidence regions, such constraints may encourage different views or branches to reproduce similar errors, thereby stabilizing in correct predictions. In addition, contrastive prototype-based methods [29]–[32] enhance feature discriminability, but their prototypes or contrastive pairs are often constructed from initial pseudo-labels. If these labels remain unreliable, feature alignment may inherit the same supervision bias. Therefore, existing methods usually improve either pseudo-label reliability or feature discriminability, while the bidirectional interaction between pseudo-label rectification and feature-space reorganization remains insufficiently explored.

To address these challenges, we propose a Diffusion-driven Hard-Soft Prototype Contrastive Learning framework (DHPCL), as illustrated in Fig. 1(b). Instead of merely filtering unreliable pseudo-labels, DHPCL actively rectifies noisy supervision and uses the diffusion-refined pseudo-labels to reorganize the feature space. Specifically, we design a Discriminator-Guided Diffusion Rectification (DGDR) module for confidence-aware pseudo-label correction. Usually, pseudo-label errors in medical image segmentation are usually structured rather than isolated, including boundary ambiguity, regional discontinuity, and anatomical shape distortion. Diffusion models can provide a more suitable iterative denoising mechanism for progressive structure restoration than one-shot filtering, morphological post-processing, or generic generation strategies. Therefore, we leverage discriminator-derived pixel-wise confidence to guide the diffusion process, applying stronger reconstruction to unreliable regions while preserving reliable anatomical structures. Furthermore, to compensate for possible over-smoothing during diffusion refinement, a Local Detail Enhancement (LDE) mechanism is introduced to preserve fine anatomical boundaries. Based on the rectified pseudo-labels, we further design a Hard-Soft Harmonized Prototype Contrast (HSH-PC) module, which constructs dynamic class prototypes from diffusion-refined pseudo-labels and applies hard constraints to high-confidence pixels together with soft distributional guidance for ambiguous pixels. In this way, DHPCL establishes a confidence-guided label-feature collaborative optimization mechanism: DGDR improves the reliability of pseudo-label supervision, while HSH-PC reorganizes the feature space according to the reliability of the rectified labels, thereby breaking the error accumulation cycle shown in Fig. 1(a).

The main contributions of this work can be summarized as follows.

- Unlike existing semi-supervised medical image segmentation approaches, we propose DHPCL, a diffusion-driven framework that rectifies noisy pseudo-labels generated by the segmentation network and feeds the refined signals back for closed-loop optimization, thereby improving supervision quality and segmentation performance.
- We design DGDR, which leverages region-wise confidence from a discriminator and LDE in the diffusion encoder to adaptively improve pseudo-label quality while preserving structural consistency.
- We design HSH-PC, where class prototypes are dynamically constructed from both diffusion-corrected and predicted pseudo-labels. Hard constraints from high-confidence regions are enforced while low-confidence regions are softly associated, promoting intra-class compactness and inter-class separability in the feature space.
- Extensive experiments on the ACDC, MS-CMRSeg, and Left Atrium (LA) datasets validate the effectiveness of DHPCL and demonstrate its superiority over existing semi-supervised methods.

II. Related Work

A. Semi-Supervised Medical Image Segmentation

1) Pseudo-label Learning: Pseudo-label learning is widely used in semi-supervised medical image segmentation. Early

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

methods, such as the influential work by Lee et al. [33], directly assigned the most probable class, but this strategy often results in substantial noise due to limited labeled samples. Some subsequent studies introduced reliability-based filtering to suppress low-quality pseudo-labels, including confidence and uncertainty estimation by Rizve et al. [19] and cross-subnetwork agreement filtering by Su et al. [12]. Although these approaches alleviate label noise, confidence-driven filtering alone offers limited gains in feature discriminability. To enhance representational robustness, several works integrated contrastive learning into pseudo-label learning frameworks. Basak et al. [34] introduced class-level contrastive supervision to strengthen inter-class separability, while Miao et al. [35] combined dual-network fusion with pixel-wise contrastive learning to improve structural coherence. Despite their effectiveness, these approaches still rely on static pseudo-labels, which cannot adapt to the varying reliability of predictions throughout training. To improve non-stationary pseudo-label quality, dynamic refinement strategies have been developed. For example, Lu et al. [20] proposed hierarchical stage-wise pseudo-label correction, and Ma et al. [21] introduced mutual-confidence fusion between foundation models and traditional models. More recent efforts further leverage multi-expert consistency [22] and boundary-aware contrastive learning with uncertainty-weighted aggregation [36] to produce higher-quality pseudo-labels.

Nevertheless, existing methods still struggle to jointly achieve dynamic adaptability, structural fidelity, and robustness to accumulated noise, motivating the need for more effective pseudo-label refinement mechanisms.

2) Consistency Regularization: Consistency regularization assumes prediction invariance under perturbations. Early work by Sohn et al. [37] enforced output consistency using cross-entropy loss, laying the foundation for subsequent approaches. To enhance robustness, Wu et al. [23] introduced adversarial perturbations via a Dice-based noise generator, and Bai et al. [38] adopted a bidirectional copy-paste scheme within a Mean Teacher framework, though it is susceptible to random paste regions and semantic mismatch. Chi et al. [39] mitigated these issues with an adaptive bidirectional shift mechanism that adjusts patch-level perturbations based on confidence. Further work explored richer perturbation and broader spatial scopes, such as dual perturbations [40] and frequency-domain mixing [24]. Multi-model collaboration has also been investigated: Song et al. [25] employed a dual-student, single-teacher design for bias correction, while Zhao et al. [26] proposed a multi-teacher, single-student framework with multi-loss optimization. To reduce the impact of noisy pseudo-labels, Zhang et al. [27] introduced a three-branch consistency model with evidence augmentation and uncertainty weighting, and Zeng et al. [28] incorporated differential feature feedback to adaptively strengthen consistency. Yang et al. [41] proposed a weak-to-strong feature perturbation strategy coupled with edge-aware contrastive learning, improving feature discriminability and edge segmentation accuracy. Despite these advances, reliance on pseudo-label quality and localized perturbations limits feature optimization in complex or low-confidence regions, hindering robustness and generalization.

Unlike these methods, DHPCL leverages DGDR and HSHPC to jointly refine noisy pseudo-labels and align features, thereby enhancing boundary correction, segmentation stability, and feature discriminability.

B. Diffusion Model

In recent years, diffusion models have demonstrated remarkable potential in generative representation learning and structure refinement for medical image segmentation. Nevertheless, semi-supervised segmentation still faces challenges due to limited labeled data and noisy pseudo-labels. Wu et al. [42] first introduced probabilistic diffusion models to this field, which were later enhanced with anchor conditioning, uncertainty-aware spatial attention, and spectral-domain Transformers to achieve variance control and semantic alignment [43]. Building on these advances, diffusion models have been applied to alleviate annotation scarcity and improve pseudo-label quality. Specifically, Yao et al. [44] employed latent diffusion models to generate large-scale unlabeled ultrasound images for semi-supervised training, while Liu et al. [45] used diffusion mechanisms to align pseudo-label distributions and reduce supervisory noise. Chen et al. [46] leveraged diffusion-based multi-scale feature extraction combined with a dual-branch image consistency strategy to enforce multi-level consistency between weak and strong perturbations. Zhang et al. [47] further integrated feature decomposition and distillation via variational autoencoders with conditional mixed diffusion models to synthesize labeled target-domain images, addressing annotation scarcity for segmentation network training. Collectively, these studies indicate that diffusion models can enhance data quality, refine pseudo-labels, and strengthen feature consistency, thereby improving semi-supervised medical image segmentation.

However, existing methods still lack dynamic, pixel-level confidence guidance during generation or refinement, often leading to insufficient handling of low-confidence regions and failing to simultaneously preserve boundary precision and structural consistency. This limitation motivates the introduction of our DGDR for adaptive pseudo-label denoising and structural refinement.

C. Prototype Contrastive Learning

Prototype-based contrastive learning has proven effective in enhancing feature discriminability for medical image segmentation by enforcing class-centered constraints. Inspired by this idea, Wang et al. [48] combined self-reference regularization with contrastive learning to improve intra-class compactness and inter-class separability. To better exploit prototype information, Song et al. [49] extended this paradigm by constructing multi-sample prototypes and incorporating semantic guidance, thereby improving prototype representations. Progressing toward semi-supervised settings, Wu et al. [29] adapted prototype contrastive learning through a prototype-aware loss that aggregates pixel features towards their class prototypes and repels them from others under pseudo-label supervision. Recognizing the limitations of fixed prototypes,

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

Liang et al. [30] further introduced adaptive prototype modeling by integrating global prototypes with confidence-weighted instance-specific prototypes to capture intra-class variations more effectively. To improve boundary representations, Wang et al. [31] expanded prototype learning with boundary-aware dual prototypes, jointly improving feature consistency and boundary discrimination. More recently, Lv et al. [32] incorporated prototype contrastive learning into the Mean Teacher framework, proposing a clustering-guided prototype mechanism to address class imbalance in semi-supervised segmentation.

Unlike static-prototype methods, HSH-PC constructs adaptive prototypes from diffusion-optimized pseudo-labels and uses Hard-Soft constraints for pixel-prototype alignment, improving noise suppression, boundary smoothing, and feature consistency.

III. Method

A. Overall Framework

The overall framework of the proposed DHPCL is shown in Fig. 2, consisting of a backbone segmentation network, DGDR, and HSH-PC.

We denote the labeled and unlabeled subsets as $D_L = \{(X_i, Y_i)\}_{i=1}^{N}$ and $D_U = \{X_j^u\}_{j=1}^{M}$, respectively, and $N \ll M$. For labeled data, weakly augmented images x_w^L are fed into the segmentation network $\mathcal{F}_\phi$, and reliable predictions selected by the confidence threshold τ are combined with the ground truth for supervised learning. For unlabeled data, weakly and strongly augmented views (x_w^U and x_s^U) are used to generate pseudo-labels Y_w^U and Y_s^U, on which a consistency loss is imposed to improve generalization.

In DGDR, a discriminator guides latent diffusion-based pseudo-label refinement. Regional confidence maps estimated from network predictions and preliminary pseudo-labels steer low-step diffusion with LDE to restore structural details and yield high-confidence pseudo-labels.

In HSH-PC, class prototypes are derived from diffusion-refined pseudo-labels and network predictions. Hard constraints enforce pixel-prototype assignments, while soft guidance preserves distribution consistency, jointly enhancing intra-class compactness and inter-class separability.

B. Discriminator-Guided Diffusion Rectification

To mitigate noise, boundary ambiguity, and structural distortions in pseudo-labels, we design DGDR, as shown in Fig. 2(b). It consists of pseudo-label quality assessment, adaptive diffusion control, and LDE.

1) Pseudo-Label Quality Assessment: Pseudo-label reliability varies spatially, so unguided refinement may overcorrect reliable regions while undercorrecting noisy ones. Therefore, DGDR uses a discriminator to estimate pixel-level reliability and generate a confidence map for diffusion-based rectification. For labeled data D_L, GT and predictions Y_w^L are one-hot encoded and fed into D_θ, where only pixels with confidence above τ are retained, producing the confidence map $\mathcal{C}_L(i,j)$:

$$\mathcal{C}_L(i,j) = \sigma(D_\theta(Y(i,j))), \tag{1}$$

where $Y \in GT, Y_w^L$. The discriminator output is normalized by the Sigmoid function $\sigma(\cdot)$ to obtain $\mathcal{C}_L(i,j)$, where values close to 1 indicate ground truth-like pixels and values close to 0 indicate pseudo-label pixels.

For unlabeled data D_U, pseudo-label reliability is estimated by comparing the refined weak prediction $\hat{Y}_w^U$ with the strong prediction Y_s^U. Specifically, $\hat{Y}_w^U$ is obtained by applying the subsequent adaptive diffusion rectification process to Y_w^U. After spatial alignment and thresholding, both predictions are fed into the discriminator to obtain $\mathcal{C}_U(i,j)$:

$$\mathcal{C}_U(i,j) = \sigma(D_\theta(\hat{Y}_w^U(i,j), Y_s^U(i,j))), \tag{2}$$

where $\mathcal{C}_U(i,j)$ represents the confidence of the pseudo-label in pixel (i,j) for unlabeled samples, with higher values indicating greater reliability of the corresponding pseudo-label.

Jointly trained on labeled and unlabeled data, the discriminator learns pseudo-label quality distributions and provides pixel-wise confidence for subsequent diffusion rectification.

2) Adaptive Diffusion Control: To avoid uniform diffusion over-modifying reliable regions or under-correcting noisy ones, we define an adaptive refinement metric Γ using the discriminator-derived confidence map $\mathcal{C} \in \mathcal{C}_L, \mathcal{C}_U$ and pixel-wise Dice loss to control diffusion strength.

First, confidence values are converted into adaptive weights for diffusion control:

$$w_{\text{adv}} = \Lambda(1 - \mathcal{C}, 0.5, 1), \tag{3}$$

where Λ truncates $1 - \mathcal{C}$ to [0.5, 1], emphasizing low-confidence pixels while preserving reliable regions. The pixel-wise Dice loss $\mathcal{L}_{Dice}$ is further incorporated to define Γ:

$$\Gamma = [s_{base} + (s_{max} - s_{base}) \cdot w_{\text{adv}}] \cdot \mathcal{L}_{Dice}, \tag{4}$$

where s_{base} and s_{max} denote the basic and maximum calibration coefficients, respectively.

To enable pixel-wise adaptive denoising, Γ is encoded by sinusoidal positional encoding and two Swish-activated linear layers to obtain e_Γ:

$$e_\Gamma = \text{Linear}_2\big(\text{Swish}(\text{Linear}_1(PE(\Gamma)))\big). \tag{5}$$

Subsequently, e_Γ is linearly projected, then fused with the diffusion feature map F:

$$F' = F + \text{Proj}(e_\Gamma). \tag{6}$$

Thus, the diffusion model uses t to control global noise evolution and e_Γ to modulate local pixel-wise refinement.

In the forward diffusion process, given the original image $\mathbf{x}_0$, its noisy counterpart at time step t, denoted by $\mathbf{x}_t$, can be formulated as:

$$x_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \tag{7}$$

where $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$, and ϵ follows the standard normal distribution, i.e. $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

In the reverse denoising process, the diffusion model progressively restores the sample under the combined guidance of the conditional input and F':

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\Big(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\,\epsilon_\theta(\mathbf{x}_t, t, F')\Big), \tag{8}$$

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

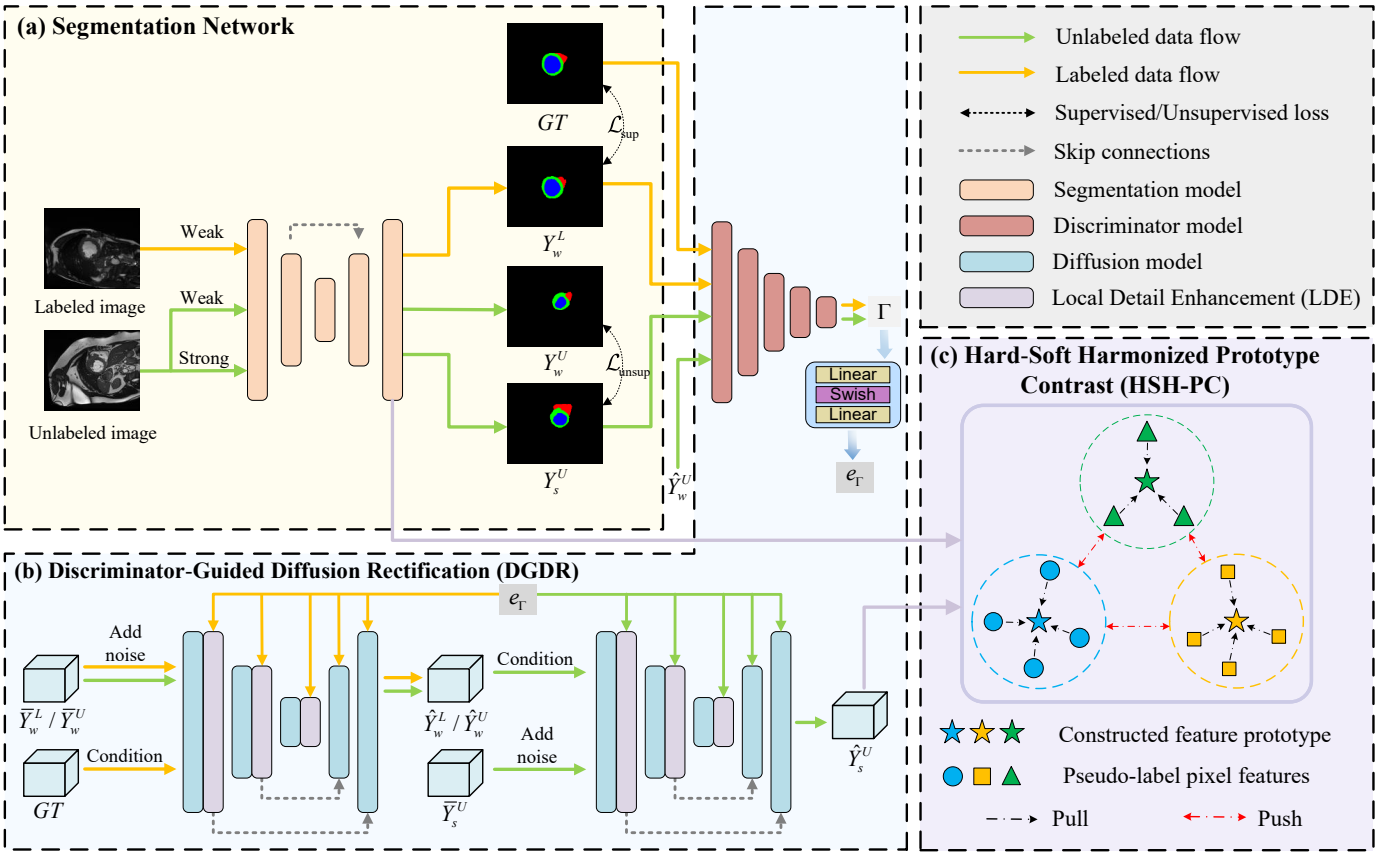


Fig. 2. Overall architecture of the proposed DHPCL. DHPCL consists of three primary components. (a) Segmentation network, responsible for extracting semantic features from input images and generating initial predictions. (b) DGDR, which utilizes discriminator-derived confidence to guide the diffusion-based refinement of pseudo-label boundaries and structural consistency. (c) HSH-PC, which constructs class prototypes from the refined pseudo-labels and performs pixel-prototype contrastive learning to strengthen feature discriminability and semantic consistency.

where ϵ_θ predicts residual noise for stepwise denoising. The timestep t controls global noise dynamics, while e_Γ within F' adaptively modulates pixel-wise refinement for fine-grained pseudo-label optimization.

DGDR adopts hierarchical progressive rectification to prevent noise amplification: weakly augmented pseudo-labels are first refined with ground truth guidance, and the corrected labels then guide the refinement of strongly augmented pseudo-labels. This process includes two steps:

(1) Weak Pseudo-Label Refinement. For labeled data, the weak prediction Y_w^L is encoded and noised, with the ground truth serving as the clean condition for reverse denoising. The supervised rectification loss is defined as

$$\mathcal{L}_{\text{weak}} = \frac{1}{|D_L|} \sum_{(X_i, Y_i)} \mathbb{E}_{t\sim U(1,T)} \left[\left\| \epsilon_\theta \left(\mathbf{z}_t^i, t, F' \right) - \mathbf{z}_{t-1}^i \right\|_2^2 \right], \tag{9}$$

where $\mathbf{z}_t^i$ denotes the noised latent representation of Y_w^L at diffusion step t, and $\mathbf{z}_{t-1}^i$ its latent state in the preceding step.

(2) Strong Pseudo-Label Refinement. For unlabeled data, the rectifier first performs reverse denoising on the noised latent representation of Y_w^U to obtain $\hat{Y}_w^U$. Since neither ground truth nor confidence guidance is available at this stage, $\text{Proj}(e_\Gamma)$ is set to 0, yielding $F' = F$. Subsequently, $\hat{Y}_w^U$ is paired with Y_s^U in Eq. (2) to compute $\mathcal{C}_U$ and then serves as the condition, while $\mathcal{C}_U$ guides the diffusion model to further rectify the strong prediction and obtain $\hat{Y}_s^U$. The strong pseudo-label rectification loss is defined as

$$\mathcal{L}_{\text{strong}} = \frac{1}{|D_U|} \sum_{X_j \in D_U} \mathbb{E}_{t\sim U(1,T)} \left[\left\| \epsilon_\theta \left(\mathbf{z}_t^j, t, F' \right) - \hat{\mathbf{z}}_{t-1}^j \right\|_2^2 \right], \tag{10}$$

where $\mathbf{z}_t^j$ is the latent representation of Y_s^U at step t, and $\hat{\mathbf{z}}_{t-1}^j$ denotes the weak pseudo-label refined latent state at step $t-1$.

3) Local Detail Enhancement: To prevent conventional convolutions from over-smoothing fine structures, we design LDE module module after the diffusion encoder convolutions for fine-grained pseudo-label restoration.

Specifically, the window size and stride are adaptively set according to the input feature map size, and $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ is divided into multiple overlapping windows:

$$\mathbf{F} \to \left\{ \mathbf{F}_p \mid \mathbf{F}_p \in \mathbb{R}^{B\times C\times ps\times ps},\ p = 1, \ldots, P \right\}, \tag{11}$$

where P denotes the total number of windows, $\mathbf{F}_p$ represents the feature submap of the p-th local window, and ps indicates the size of each local window, which is calculated as follows:

$$ps = \begin{cases} \min(H, W), & \text{if } \min(H, W) \le K \\ K, & \text{Otherwise} \end{cases}, \tag{12}$$

where K denotes the maximum patch size, H and W represent the height and width of the feature map, respectively. The stride is computed as follows:

$$step = \max(1, \lfloor ps/2 \rfloor). \tag{13}$$

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

Subsequently, multi-head self-attention is then performed in each window and fused with residual and depthwise separable feed-forward operations to obtain $\tilde{\mathbf{F}}_p$:

$$\tilde{\mathbf{F}}_p = \mathbf{F}_p + L_2(G(L_1(\mathbf{X})) + \mathrm{DWConv}(G(L_1(\mathbf{X})))), \quad (14)$$

where L_1 and L_2 are linear projections, $G(\cdot)$ is GELU, $\mathrm{DWConv}(\cdot)$ is depthwise separable convolution. The residual feature $\mathbf{X}$ is computed as

$$\mathbf{X} = \mathrm{Attention}(\mathbf{F}_p) + \mathbf{F}_p. \quad (15)$$

After processing all windows, the enhanced local window features are reintegrated into the original feature map through averaging over the overlapping regions,

$$\tilde{\mathbf{F}}(i,j) = \frac{1}{|\Omega(i,j)|} \sum_{p:(i,j)\in\mathbf{F}_p} \tilde{\mathbf{F}}_p(i,j), \quad (16)$$

where $\Omega(i,j)$ denotes the set of windows that contain the pixel (i,j).

Through LDE, the diffusion model captures fine-grained window-level features and cooperates with discriminator-guided adaptive weights to improve pseudo-label restoration and segmentation accuracy.

The overall processing of labeled and unlabeled data in DGDR is summarized in Algorithm 1.

Algorithm 1 Overall Processing of DGDR

Input: Labeled data D_L, unlabeled data D_U
Output: Refined strong pseudo-label $\hat{Y}_s^U$

for each labeled and unlabeled mini-batch **do**
  Generate Y_w^L, Y_w^U, and Y_s^U using the Segmentation Network;
  // Supervised learning of the diffusion rectifier
  if the processing sample $\in D_L$ **then**
    Compute $\mathcal{C}_L$ using Eq. (1);
    Train the diffusion rectifier on Y_w^L with ground truth and $\mathcal{C}_L$ guidance using Eqs. (3)-(9);
  end if
  // Hierarchical unlabeled pseudo-label rectification
  if the processing sample $\in D_U$ **then**
    Obtain $\hat{Y}_w^U$ by applying the learned rectifier to Y_w^U;
    Compute $\mathcal{C}_U$ from $(\hat{Y}_w^U, Y_s^U)$ using Eq. (2);
    Construct the guided feature F_U' using Eqs. (3)-(6);
    Add noise to Y_s^U using Eq. (7);
    Obtain $\hat{Y}_s^U$ by denoising and rectifying the resulting latent using Eq. (8);
    Compute $\mathcal{L}_{\mathrm{strong}}$ using Eq. (10);
  end if
  Update DGDR using the corresponding discriminator and diffusion rectification losses;
end for

C. Hard-Soft Harmonized Prototype Contrast

Although DGDR improves pseudo-label structural coherence, label supervision alone is insufficient to ensure feature compactness and separability. Therefore, we design HSH-PC to enhance feature discrimination and global consistency through prototype construction, EMA-based updating, and pixel-prototype contrastive learning, as shown in Fig. 3.

1) Prototype Construction: First, the decoder feature map $F \in \mathbb{R}^{B\times C\times H\times W}$ is extracted for prototype construction, and the diffusion-refined pseudo-labels $\hat{Y}_s^U$ are resized to $\hat{y} \in \mathbb{R}^{B\times C\times H\times W}$ for pixel-wise alignment. For class c, the instantaneous prototype is defined as

$$p_c^{(t)} = \frac{\sum_{i,j} F_{\{b:,(i,j)\}} \cdot \mathbb{I}[\hat{y}(i,j)=c]}{\sum_{i,j} \mathbb{I}[\hat{y}(i,j)=c] + \epsilon'}, \quad (17)$$

where $F_{\{b,:,i,j\}}$ denotes the feature vector at the pixel location (i,j) of the b-th sample. $\mathbb{I}[\cdot]$ represents the indicator function and ϵ' is a small constant added to prevent division by zero.

2) EMA-based Prototype Updating: Global prototypes are updated by EMA to reduce batch-level fluctuations. For class c, the prototype is initialized as $p_c^{(t)}$ at its first appearance and otherwise updated recursively as

$$\bar{p}_c^{(t)} = \alpha \bar{p}_c^{(t-1)} + (1-\alpha) p_c^{(t)}, \quad (18)$$

where α is the momentum coefficient. The EMA update stabilizes prototypes by integrating historical and current features without gradient backpropagation, and $\bar{p}_c^{(t)}$ serves as the class anchor in HSH-PC.

3) Pixel–Prototype Contrastive Learning: Pixel-prototype contrastive learning further alleviates intra-class dispersion and inter-class confusion by using hard and soft constraints to align pixels with same-class prototypes and separate them from others.

(1) Similarity Computation. The pixel feature vector f_i and EMA prototype vector $\bar{p}_j$ are first L2-normalized, and their similarity is computed by cosine similarity as

$$\Phi_{i,j} = \mathrm{Softmax}\left(\frac{f_i \cdot \bar{p}_j}{\|f_i\|_2 \|\bar{p}_j\|_2} / \tau'\right), \quad (19)$$

where τ' is the temperature parameter controlling the sharpness of the Softmax distribution. This similarity measures pixel-prototype alignment and supports subsequent positive-negative contrastive learning.

(2) Positive-Negative Matching Construction. Instead of applying a uniform matching rule to all pixels, we construct a confidence-routed target $\Psi_{i,j}$ that adaptively assigns each pixel either a hard one-hot correspondence or a soft distributional correspondence with the class prototypes, depending on its prediction confidence:

$$\Psi_{i,j} = \begin{cases} \mathbb{I}[y_i = c_j], & \mathrm{conf}_i \geq \delta \\ \dfrac{\exp(\hat{p}_i^j/\tau_s)}{\sum_k \exp(\hat{p}_i^k/\tau_s)}, & \mathrm{conf}_i < \delta \end{cases} \quad (20)$$

where y_i is the flattened class index of $\hat{y}$, c_j denotes the j-th class, and $\hat{p}_i \in \mathbb{R}^C$ denotes the pixel-wise softmax probability vector of the diffusion-refined pseudo-label map $\hat{y}$ at pixel i with confidence $\mathrm{conf}_i = \max_c \hat{p}_i^c$. Here $\delta = 0.7$ routes each pixel to a hard or soft regime rather than discarding it, and τ_s controls the sharpness of the soft distribution.

The final pixel-prototype contrastive loss is defined as

$$\mathcal{L}_{\mathrm{proto}} = -\frac{1}{|\Omega|} \sum_{i\in\Omega} \sum_j \Psi_{i,j} \cdot \log \frac{\exp(\Phi_{i,j})}{\sum_k \exp(\Phi_{i,k})}, \quad (21)$$

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

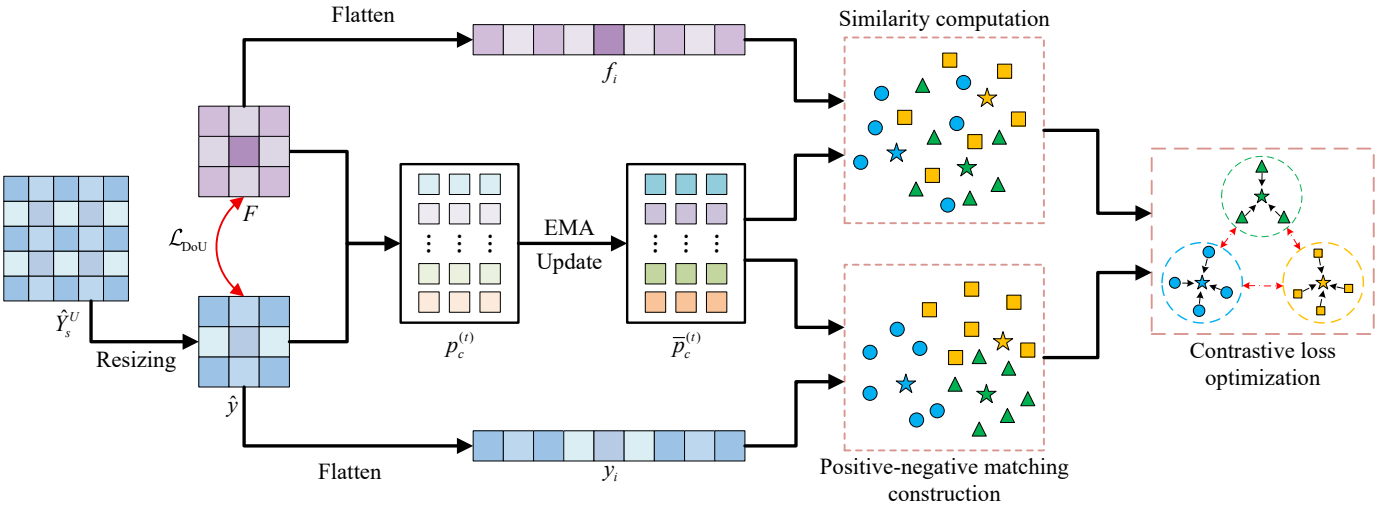


Fig. 3. Overview of the HSH-PC module. Decoder features $\boldsymbol{F}$ and diffusion corrected pseudo-labels $\hat{\boldsymbol{y}}$ are used to construct class prototypes, which are then stabilized through EMA updating. Pixel features are matched with these prototypes to form hard and soft associations, and contrastive optimization is applied to enforce discriminative feature alignment.

where Ω denotes the full set of pixels participating in contrastive optimization (comprising both high- and low-confidence pixels). This loss unifies hard and soft supervision within a single objective, jointly enhancing intra-class compactness and inter-class separability.

Inspired by [50], we combine the boundary loss $\mathcal{L}_{\text{DoU}}$ with the prototype contrastive loss in a weighted manner to obtain the final contrastive loss function

$$\mathcal{L}_{\text{p2p}} = \mathcal{L}_{\text{DoU}} + \lambda_{\text{p2p}}\mathcal{L}_{\text{proto}}\,, \tag{22}$$

where λ_{p2p} balances the prototype contrastive loss to improve feature discriminability and generalization.

D. Loss Function

The overall loss comprises supervised, unsupervised consistency, discriminator adversarial, diffusion rectification, and pixel–prototype contrastive losses, formulated as

$$\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda_1\mathcal{L}_{\text{unsup}} + \lambda_2(\mathcal{L}_L + \mathcal{L}_U) \\ + \lambda_3(\mathcal{L}_{\text{weak}} + \mathcal{L}_{\text{strong}}) + \lambda_4\mathcal{L}_{\text{p2p}},\end{aligned} \tag{23}$$

where $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0.1$.

The supervised loss $\mathcal{L}_{\text{sup}}$ is calculated based on the fully annotated samples as

$$\mathcal{L}_{\text{sup}} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathcal{L}_{\text{Dice}}(Y_w^L, GT) + \lambda_{CE}\mathcal{L}_{CE}(Y_w^L, GT)\right), \tag{24}$$

where N is the number of labeled samples, Y_w^L the weakly augmented predictions, and λ_{CE} the weighting coefficient.

The unsupervised consistency loss $\mathcal{L}_{\text{unsup}}$ enforces agreement between weakly and strongly augmented unlabeled predictions to improve generalization,

$$\mathcal{L}_{\text{unsup}} = \frac{1}{M}\sum_{j=1}^{M}\|\text{Softmax}(Y_w^U) - \text{Softmax}(Y_s^U)\|_2^2, \tag{25}$$

where M denotes the number of unlabeled samples.

The optimization objective of the discriminator is composed of the loss for labeled data, $\mathcal{L}_L$, and the loss for unlabeled data, $\mathcal{L}_U$. Labeled data loss, $\mathcal{L}_L$, is defined as

$$\begin{aligned}\mathcal{L}_L = -\frac{1}{B\cdot H\cdot W}\sum_{b=1}^{B}\sum_{i=1}^{H}\sum_{j=1}^{W}\Big[y_{i,j}\log\mathcal{C}_L(i,j) \\ + (1-y_{i,j})\log(1-\mathcal{C}_L(i,j))\Big],\end{aligned} \tag{26}$$

where $y_{i,j}$ = 1/0 denotes ground truth/network-predicted pixels, and $z_{i,j}$ measures pixel-wise consistency between weak and strong augmentation predictions for unlabeled data,

$$z_{i,j} = 1 - \left\|\text{Softmax}\left(Y_w^U(i,j)\right) - \text{Softmax}\left(Y_s^U(i,j)\right)\right\|_2. \tag{27}$$

Higher $z_{i,j}$ values indicate more reliable pixel predictions, while lower values denote uncertainty. On unlabeled data, the discriminator leverages $z_{i,j}$ in a binary cross-entropy loss to smooth training and provide progressive supervision.

$$\begin{aligned}\mathcal{L}_U = -\frac{1}{B\cdot H\cdot W}\sum_{b=1}^{B}\sum_{i=1}^{H}\sum_{j=1}^{W}\Big[z_{i,j}\log\mathcal{C}_U(i,j) \\ + (1-z_{i,j})\log(1-\mathcal{C}_U(i,j))\Big].\end{aligned} \tag{28}$$

IV. Experiments

A. Datasets

We evaluated DHPCL on three public datasets, including ACDC [51], MS-CMRSeg [52], [53], and Left Atrium (LA) Dataset [54]. The first two are processed as 2D slice-based segmentation tasks, while LA is used for 3D volumetric segmentation.

ACDC Dataset [51]: This dataset contains of cine cardiac MRI scans from 100 patients, with annotations for the right ventricle, left ventricle, myocardium, and background. Referring to [20], [55], we split it into 70 training, 10 validation, and 20 testing cases.

MS-CMRSeg Dataset [52], [53]: This dataset contains multi-sequence cardiac MRI scans from 45 patients with

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

cardiomyopathy. The dataset is divided into 30 cases for training, 5 cases for validation, and 10 cases for testing.

LA Dataset [54]: This dataset contains 100 3D gadolinium-enhanced magnetic resonance imaging (GE-MRI) scans with manual annotations of the left atrium. Following [12], we adopted a fixed split, with 80 samples for training and the remaining 20 samples for testing.

To ensure the reliability and consistency of our experimental results, we conducted five-fold cross-validation on the ACDC and LA datasets, and four-fold cross-validation on the MS-CMRSeg dataset. For all evaluations, the final results are recorded and reported as the mean±standard deviation.

B. Implementation Details

1) Training: We implemented DHPCL in PyTorch and conducted all experiments on a single NVIDIA GeForce RTX 3090 GPU. For ACDC and MS-CMRSeg, each case contains multiple 2D slices, and labeled data are sampled at the case level: once a case is selected, all its slices are used as labeled data, while the remaining training cases are treated as unlabeled data. All 2D slices are resized to 256×256, and a 2D U-Net is used as the segmentation backbone. The discriminator is implemented as a five-layer 2D convolutional network, with the first four layers using 4×4 kernels (stride=2, and padding=1) for progressive downsampling, and the final layer generating pixel-wise confidence maps. For the LA dataset, each 3D volume is preprocessed to $112 \times 112 \times 80$ [12]. We adopted a 3D V-Net backbone and constructed a corresponding 3D discriminator by replacing 2D convolutions with 3D convolutions to estimate voxel-wise confidence maps.

All models are optimized with Adam using an initial learning rate of 0.001 and trained for 30,000, 30,000, and 15,000 iterations on ACDC, MS-CMRSeg, and LA, respectively. The batch size is 12 for 2D tasks, including 6 labeled and 6 unlabeled slices, and 4 for the 3D task, including 2 labeled and 2 unlabeled volumes. Labeled samples are augmented with weak transformations, including random flipping and small-angle rotation, while unlabeled samples are augmented into weak and strong views, with the strong view further using color jittering and Gaussian blurring. Based on validation performance, τ, λ_{p2p}, and the numbers of diffusion and reverse sampling steps are set to 0.85, 0.5, 20, and 4, respectively, with a cosine schedule used for noise injection.

2) Testing: During inference, only the trained segmentation network is used to generate the final segmentation prediction. The DGDR and HSH-PC module are used only during training and are not involved in the inference stage.

3) Evaluation Metrics: We evaluate segmentation performance using four metrics: Dice Similarity Coefficient (DSC), Jaccard Index, 95% Hausdorff Distance (HD95), and Average Surface Distance (ASD). During training, the checkpoint with the best validation DSC is selected for final testing, and all hyperparameters are tuned only on the validation set.

C. Comparison With State-of-The-Art Methods

To comprehensively evaluate the effectiveness of DHPCL, we conducted comparative experiments on the ACDC, MS-CMRSeg, and LA datasets. For the ACDC and MS-CMRSeg datasets, DHPCL is compared with a fully supervised baseline, nn-UNet [56], and several state-of-the-art semi-supervised segmentation methods, including SSNet [23], PatchCL [34], ICL [57], BCP [38], DCNet [58], DPLR [20], AD-MT [26], ABD [39], SDCL [25], CGS [22], β-FFT [24], UCSeg [16], and HASSL [59]. For the LA dataset, DHPCL is further compared with representative semi-supervised 3D segmentation methods, including PMT [15], Mutual [12], UCSeg [16], UnCo [60], and HASSL [59].

1) ACDC Dataset: Table I presents the quantitative results on ACDC with 5% labeled data. The fully supervised nn-UNet achieves DSC score of 91.58% and HD95 of 1.46 mm, serving as the upper bound with sufficient annotations. In contrast, semi-supervised methods such as SSNet and DCNet show clear degradation due to pseudo-label noise and anatomical variability. Although BCP, DPLR, AD-MT, and SDCL improve performance through data augmentation or consistency learning, they still struggle with boundary delineation in low-confidence regions. DHPCL achieves the best overall performance by leveraging DGDR and HSH-PC, surpassing the second-best method by 1.04% in DSC and 1.83% in Jaccard, while maintaining competitive HD95 and ASD, demonstrating its effectiveness in low-label scenarios.

Table II presents the evaluation results under the 10% labeled data setting. As the proportion of labeled data increases, overall performance improves further. DHPCL achieves DSC, Jaccard, HD95, and ASD scores of 90.95%, 83.74%, 1.32 mm, and 0.34 mm, respectively, ranking first in terms of all metrics. Compared to BCP, DHPCL yields absolute improvements of 2.50%, 3.07%, 2.69 mm, and 0.80 mm on four metrics, respectively. Moreover, even compared with the fully supervised nn-UNet, DHPCL still surpasses it in terms of HD95 and ASD by 0.14 mm and 0.15 mm, respectively, demonstrating that the semi-supervised learning strategy not only compensates for the limited availability of annotations, but also improves boundary precision beyond the fully supervised baseline.

Table III presents the results under the 20% labeled data. DHPCL continues to exhibit a distinct advantage, achieving an average DSC of 91.21% and a Jaccard index of 84.30%, both ranking among the best across all competing methods.

Fig. 4 presents a qualitative comparison on the ACDC dataset with 10% labeled data. DHPCL generates smoother boundaries and more coherent RV, Myo, and LV structures. In contrast, SSNet, PatchCL, and ICL suffer from pseudo-label noise, while BCP and DCNet still produce jagged local details despite improved contours. DPLR, AD-MT, ABD, and SDCL show local omissions in ambiguous regions, and CGS and β-FFT still exhibit minor boundary deviations and gaps. Overall, DHPCL better preserves fine boundary details and global structural consistency, demonstrating superior segmentation performance.

2) MS-CMRSeg Dataset: Table IV summarizes the quantitative results on the MS-CMRSeg dataset with 20% labeled data. DHPCL achieves DSC, Jaccard, HD95, and ASD values of 88.84%, 80.31%, 4.96 mm, and 1.16 mm, respectively, outperforming most competing methods. Although SDCL mitigates pseudo-label bias through dual-student discrepancy-guided correction, its reliance on prediction discrepancies limits pixel-

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

TABLE I

QUANTITATIVE RESULTS GENERATED BY DIFFERENT METHODS ON THE ACDC WITH THE 5% LABELED DATA RATIO. THE BEST VALUES ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Method	RV				Myo				LV				Mean			
	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓
nn-UNet(Nat.Met.'21) [56]	91.28±0.45	86.12±0.52	1.56±0.28	0.65±0.21	88.45±0.62	78.63±0.55	1.25±0.46	0.46±0.11	95.01±0.58	91.41±0.49	1.58±0.33	0.37±0.10	91.58±0.51	85.39±0.48	1.46±0.12	0.49±0.19
SSNet (MICCAI'22) [23]	57.75±1.65	47.58±1.72	9.41±2.85	3.66±0.68	65.21±1.71	53.57±1.66	6.79±1.79	2.25±0.53	74.48±1.63	65.01±1.13	3.52±1.63	1.14±0.25	65.81±0.90	55.39±1.68	6.57±2.74	2.35±0.58
ICL (MIDL'23) [57]	79.22±0.94	68.69±1.92	7.65±1.07	2.34±0.75	80.45±1.88	68.46±0.95	2.79±0.22	0.56±0.48	84.43±1.63	77.51±1.82	14.52±3.20	12.81±3.67	81.37±1.42	71.55±0.98	8.32±2.11	5.23±1.20
BCP (CVPR'23) [38]	85.59±1.57	74.72±1.36	5.44±1.75	1.46±0.53	83.01±1.18	72.15±1.20	8.41±2.95	2.03±0.61	90.92±1.45	85.05±1.66	15.42±3.90	3.47±0.79	86.51±1.77	77.30±1.19	9.76±2.00	2.32±0.66
DCNet (MICCAI'23) [58]	64.58±1.55	53.45±0.92	3.61±1.58	1.42±0.45	70.15±1.49	58.42±1.65	3.09±1.32	1.31±0.42	78.81±1.87	71.29±1.90	6.59±2.63	2.01±0.50	71.18±1.58	61.05±1.17	4.43±1.51	1.58±0.84
DPLR (ISBI'24) [20]	66.83±1.18	54.69±1.58	6.45±3.70	2.26±0.92	75.31±1.63	61.23±1.17	3.48±1.55	1.17±0.24	86.52±1.91	76.79±1.49	5.64±2.68	1.79±0.36	76.22±1.36	64.24±1.26	5.19±2.64	1.74±0.45
AD-MT (ECCV'24) [26]	84.13±1.50	74.54±1.55	4.17±0.83	1.38±0.42	86.75±1.58	76.07±0.62	2.71±0.70	0.76±0.38	90.68±1.62	84.96±1.56	4.38±0.74	1.45±0.41	87.19±1.13	78.52±1.27	3.75±0.82	1.19±0.39
ABD (CVPR'24) [39]	87.15±1.48	77.11±1.71	2.13±0.40	1.28±0.25	83.82±1.53	73.40±0.96	2.00±0.36	0.83±0.14	91.59±1.49	84.37±1.32	2.94±0.31	1.04±0.27	87.52±1.50	78.29±1.43	**2.06±0.47**	1.09±0.32
SDCL (MICCAI'24) [25]	86.65±2.52	78.03±1.14	2.39±1.45	0.81±0.19	84.25±1.94	72.45±1.68	**1.42±0.34**	0.72±0.18	91.01±1.71	84.92±2.03	2.86±0.91	1.22±0.24	87.30±1.52	78.48±1.51	2.16±0.79	0.60±0.21
CGS (TMI'25) [22]	84.78±2.56	73.52±1.59	6.51±2.62	1.81±0.28	83.86±1.71	74.45±1.61	2.29±1.16	0.99±0.14	91.19±1.55	83.63±1.57	**2.64±1.49**	1.09±0.13	86.61±1.66	76.96±1.49	3.81±1.37	1.30±0.17
β-FFT (CVPR'25) [24]	88.07±0.46	79.16±0.49	2.05±0.39	0.67±0.12	86.01±0.40	76.37±0.33	1.94±0.41	0.95±0.18	91.98±0.37	86.07±0.30	2.81±0.41	1.10±0.39	88.79±0.48	80.32±0.41	2.40±0.30	0.90±0.26
UCSeg (TIP'25) [16]	87.31±0.45	79.15±0.40	2.09±0.35	1.04±0.10	86.53±0.41	75.75±0.33	2.37±0.32	0.66±0.15	91.02±0.56	85.91±0.51	2.85±0.41	1.07±0.19	88.62±0.37	80.27±0.42	2.44±0.36	0.93±0.14
HASSL (PR'26) [59]	87.87±0.30	78.44±0.45	2.58±0.40	0.77±0.13	85.81±0.32	76.35±0.36	3.00±0.43	0.78±0.18	91.36±0.41	85.21±0.44	2.87±0.31	1.19±0.12	88.35±0.35	80.00±0.45	2.82±0.40	0.95±0.19
DHPCL	**89.21±0.27**	**82.13±0.31**	**1.78±0.35**	**0.59±0.10**	**87.39±0.26**	**77.00±0.32**	2.21±0.41	**0.38±0.13**	**92.90±0.33**	**87.33±0.41**	3.04±0.29	**0.73±0.14**	**89.83±0.34**	**82.15±0.47**	2.34±0.38	**0.57±0.17**

TABLE II

QUANTITATIVE RESULTS GENERATED BY DIFFERENT METHODS ON THE ACDC WITH THE 10% LABELED DATA RATIO. THE BEST VALUES ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Method	RV				Myo				LV				Mean			
	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓
nn-UNet(Nat.Met.'21) [56]	91.28±0.45	86.12±0.52	1.56±0.28	0.65±0.21	88.45±0.62	78.63±0.55	1.25±0.46	0.46±0.11	95.01±0.58	91.41±0.49	1.58±0.33	0.37±0.10	91.58±0.51	85.39±0.48	1.46±0.12	0.49±0.19
SSNet (MICCAI'22) [23]	85.47±1.32	76.25±0.87	5.42±0.98	1.26±0.23	84.21±1.56	73.40±1.12	6.38±1.51	1.37±0.26	87.75±0.94	83.44±1.65	6.25±0.94	1.49±0.32	85.81±1.41	77.70±0.79	6.02±1.39	1.37±0.41
PatchCL (CVPR'23) [34]	85.94±1.07	76.43±1.73	2.97±0.52	0.81±0.20	83.25±0.83	71.21±1.48	2.26±0.30	0.90±0.34	88.01±1.62	82.72±0.68	3.43±0.33	1.32±0.45	85.72±1.15	76.79±1.84	2.89±0.41	1.01±0.24
ICL (MIDL'23) [57]	86.38±1.44	77.01±0.91	3.02±0.42	0.93±0.26	85.01±1.27	74.70±1.53	2.09±0.47	0.74±0.19	87.37±0.76	84.61±1.38	3.86±0.35	1.12±0.21	86.25±1.69	78.77±0.85	2.99±0.40	0.93±0.32
BCP (CVPR'23) [38]	87.52±0.69	79.18±1.81	3.36±0.54	0.85±0.23	86.67±1.50	76.41±0.72	4.52±0.68	1.29±0.31	91.15±1.33	86.43±1.06	4.14±0.45	1.28±0.37	88.45±0.97	80.67±1.76	4.01±0.53	1.14±0.25
DCNet (MICCAI'23) [58]	88.95±1.18	80.27±0.89	2.41±0.29	0.75±0.14	86.45±1.71	76.78±1.24	2.33±0.21	0.57±0.22	87.40±0.81	88.26±1.59	1.71±0.31	0.55±0.18	87.60±1.47	81.77±0.65	2.15±0.31	0.62±0.13
DPLR (ISBI'24) [20]	87.51±1.63	80.13±1.02	1.72±0.33	0.52±0.11	85.99±0.78	75.35±1.87	1.97±0.25	1.06±0.34	89.80±1.22	86.33±0.93	3.62±0.49	1.33±0.27	87.77±1.54	80.60±1.31	2.44±0.32	0.97±0.20
AD-MT (ECCV'24) [26]	88.95±1.24	80.59±0.97	1.51±0.16	0.60±0.22	86.85±1.51	76.92±1.18	2.23±0.20	0.64±0.14	91.73±0.88	87.41±1.35	2.05±0.12	0.71±0.19	89.18±1.33	81.64±0.76	1.93±0.18	0.65±0.15
ABD (CVPR'24) [39]	89.81±0.75	81.94±0.62	1.93±0.23	0.59±0.11	86.01±0.77	75.67±0.35	1.42±0.25	0.74±0.24	92.09±0.40	87.34±0.43	2.69±0.29	0.68±0.17	89.64±0.48	81.65±0.41	2.01±0.14	0.67±0.10
SDCL (MICCAI'24) [25]	89.52±0.92	81.32±1.67	1.63±0.25	0.62±0.21	87.45±1.36	78.27±0.81	1.27±0.32	0.42±0.13	93.33±0.75	88.34±0.34	2.83±0.44	0.68±0.18	90.10±0.68	82.64±1.42	1.91±0.24	0.57±0.16
CGS (TMI'25) [22]	87.11±1.84	77.52±1.15	4.37±0.47	1.27±0.42	86.10±1.09	76.15±1.61	2.51±0.30	0.88±0.25	91.37±1.40	84.71±0.88	4.46±0.33	1.45±0.39	88.19±0.73	79.46±1.29	3.45±0.44	1.20±0.22
β-FFT (CVPR'25) [24]	90.25±0.21	83.24±0.18	1.51±0.20	0.49±0.14	87.80±0.16	78.10±0.34	1.82±0.13	0.55±0.21	92.80±1.57	86.80±1.32	4.11±0.16	1.23±0.11	90.28±0.40	82.71±0.69	2.48±0.18	0.76±0.23
UCSeg (TIP'25) [16]	89.42±0.52	82.17±0.96	2.63±0.31	0.58±0.19	86.93±1.18	77.24±0.72	2.95±0.24	0.64±0.15	91.94±0.71	85.92±0.45	2.28±0.27	1.31±0.20	89.43±0.66	81.78±0.82	2.62±0.22	0.84±0.24
HASSL (PR'26) [59]	89.17±1.04	81.85±1.37	1.58±0.24	0.54±0.17	86.65±0.67	76.88±0.50	2.89±0.54	0.60±0.32	91.58±0.81	85.47±0.79	3.22±0.41	1.27±0.43	89.13±0.75	81.40±0.63	2.56±0.39	0.80±0.16
DHPCL	**90.73±0.46**	**83.27±0.31**	**1.29±0.28**	**0.39±0.11**	**87.95±0.62**	**79.10±0.40**	**1.21±0.37**	**0.34±0.15**	**94.16±0.89**	**88.86±0.28**	**1.48±0.23**	**0.31±0.11**	**90.95±0.70**	**83.74±0.33**	**1.32±0.26**	**0.34±0.09**

TABLE III

QUANTITATIVE RESULTS GENERATED BY DIFFERENT METHODS ON THE ACDC WITH THE 20% LABELED DATA. THE BEST VALUES ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST ARE UNDERLINED.

Method	RV				Myo				LV				Mean			
	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓	DSC↑	Jaccard↑	HD95↓	ASD↓
nn-UNet(Nat.Met.'21) [56]	91.28±0.45	86.12±0.52	1.56±0.28	0.65±0.21	88.45±0.62	78.63±0.55	1.25±0.46	0.46±0.11	95.01±0.58	91.41±0.49	1.58±0.33	0.37±0.10	91.58±0.51	85.39±0.48	1.46±0.12	0.49±0.19
SSNet (MICCAI'22) [23]	88.21±1.26	79.93±0.85	1.64±0.18	1.25±0.22	85.45±1.53	75.02±1.17	5.16±0.15	1.42±0.24	90.15±0.94	83.29±1.62	7.81±0.21	1.96±0.13	87.92±1.75	79.41±1.38	4.85±0.12	1.59±0.20
ICL (MIDL'23) [57]	88.68±0.68	80.41±1.45	2.49±0.23	0.64±0.17	88.09±1.82	79.02±0.73	2.25±0.14	0.61±0.22	92.76±1.31	87.16±1.06	3.85±0.19	0.86±0.15	89.83±0.59	82.19±1.67	2.48±0.24	0.70±0.11
BCP (CVPR'23) [38]	88.61±1.42	80.39±0.91	1.68±0.16	0.65±0.23	88.45±1.15	78.09±1.74	3.21±0.20	0.88±0.14	92.22±0.76	87.83±1.28	2.97±0.13	1.18±0.18	89.75±1.88	82.10±0.63	2.61±0.22	0.91±0.25
DCNet (MICCAI'23) [58]	88.48±0.55	81.85±1.59	1.95±0.25	0.76±0.12	87.28±1.64	77.74±1.03	1.93±0.17	0.62±0.21	92.93±1.22	87.37±0.87	2.98±0.11	0.77±0.19	89.56±0.83	82.32±1.46	2.28±0.23	0.72±0.15
DPLR (ISBI'24) [20]	88.57±1.71	80.53±1.12	1.49±0.14	0.45±0.24	86.66±0.97	76.76±1.85	3.34±0.21	0.97±0.16	91.68±1.36	85.42±0.79	6.53±0.18	1.87±0.22	88.97±1.54	80.90±1.23	3.78±0.25	1.10±0.13
AD-MT (ECCV'24) [26]	89.08±0.62	81.15±1.68	1.33±0.15	0.40±0.20	87.55±1.81	78.17±0.95	2.67±0.22	0.55±0.17	91.71±1.08	87.99±1.34	2.83±0.13	0.70±0.23	89.45±1.47	82.44±0.72	2.27±0.19	0.55±0.11
ABD (CVPR'24) [39]	89.08±1.39	82.44±1.16	1.73±0.24	0.54±0.18	87.63±0.86	76.73±1.79	1.33±0.12	0.57±0.25	92.13±1.65	87.47±0.98	1.68±0.17	0.53±0.14	89.61±1.13	82.55±1.52	1.58±0.21	0.55±0.22
SDCL (MICCAI'24) [25]	89.02±1.77	81.18±0.67	1.55±0.19	0.40±0.15	88.51±0.53	79.66±1.41	1.15±0.23	**0.35±0.20**	92.98±1.29	87.55±1.05	2.27±0.14	0.51±0.17	90.17±0.92	82.80±1.63	1.61±0.22	0.42±0.14
CGS (TMI'25) [22]	89.54±1.45	81.75±1.19	1.28±0.16	0.42±0.13	87.53±1.84	78.09±0.75	1.31±0.20	0.61±0.21	92.81±0.78	87.26±1.57	5.02±0.24	1.10±0.18	89.96±1.69	82.37±1.01	2.48±0.12	0.71±0.23
β-FFT (CVPR'25) [24]	**91.08±1.04**	**84.06±1.72**	1.34±0.21	0.39±0.15	88.20±1.58	79.12±0.89	1.90±0.14	1.03±0.22	93.13±0.65	87.60±1.37	2.73±0.18	1.07±0.11	90.80±1.24	83.59±1.49	1.99±0.25	0.71±0.19
UCSeg (TIP'25) [16]	89.56±1.31	81.82±0.93	2.24±0.17	0.62±0.24	87.63±1.76	82.05±1.28	2.38±0.23	0.69±0.16	92.94±0.82	82.04±1.64	2.46±0.13	0.73±0.20	90.04±1.18	81.97±1.07	2.36±0.19	0.68±0.14
HASSL (PR'26) [59]	89.82±1.63	82.03±0.71	1.35±0.22	0.65±0.18	87.16±0.95	82.29±1.55	1.49±0.15	0.72±0.23	92.25±1.27	82.19±1.12	1.54±0.20	0.73±0.12	89.74±1.85	82.17±1.39	1.46±0.14	0.70±0.21
DHPCL	90.48±0.74	83.22±1.56	**1.02±0.11**	**0.29±0.07**	**88.77±1.33**	**79.97±0.81**	**1.06±0.18**	0.36±0.06	**94.23±1.09**	**89.55±1.25**	**1.19±0.23**	**0.34±0.06**	**91.21±1.67**	**84.30±1.42**	**1.09±0.15**	**0.33±0.09**

level confidence modeling in complex or ambiguous regions. In contrast, DHPCL employs discriminator-guided diffusion correction for adaptive pseudo-label refinement, while HSH-PC enhances intra-class compactness and inter-class separability. As a result, DHPCL surpasses SDCL by 1.75% and 2.65% in DSC and Jaccard, and reduces HD95 and ASD from 7.97 mm and 2.96 mm to 4.96 mm and 1.16 mm, respectively, demonstrating improved regional consistency and boundary localization.

3) LA Dataset: As shown in Table V, DHPCL demonstrates consistently superior performance on the LA dataset under both 10% and 20% labeled settings. With 10% labeled data, DHPCL obtains the best DSC, Jaccard, and ASD scores of 90.59%, 82.90%, and 1.61, respectively. Although UnCo achieves the lowest HD95, DHPCL attains the second-best HD95 with better overlap accuracy and ASD, indicating more balanced performance. With 20% labeled data, DHPCL further improves DSC and Jaccard to 91.19% and 83.87%, respectively, and achieves the best ASD of 1.51. These results show that DHPCL effectively leverages limited annotations for 3D LA segmentation by improving pseudo-label quality and feature discriminability.

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

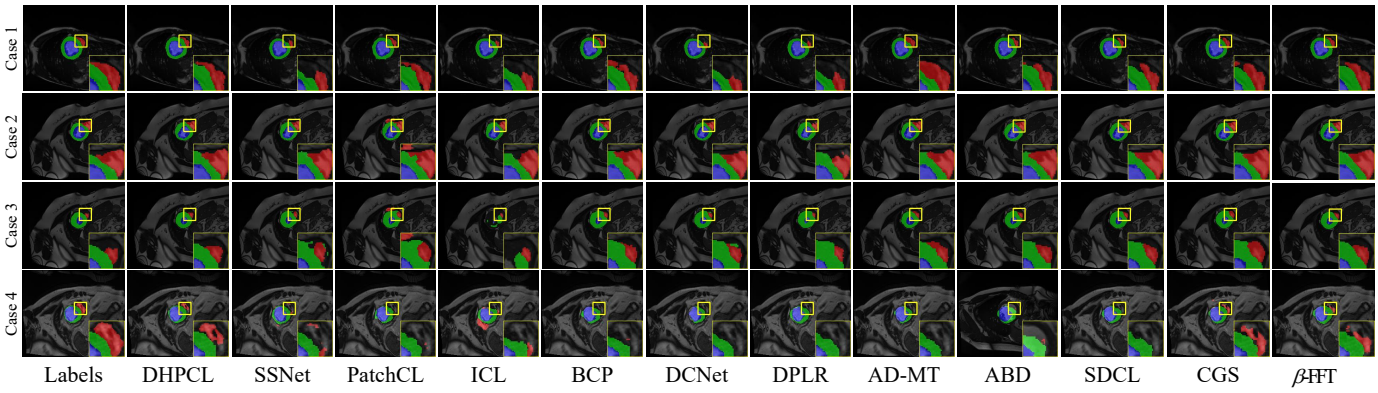


Fig. 4. Visualization results generated by different methods on the ACDC dataset.The right ventricle, myocardium, and left ventricle are shown in blue, green, and red, respectively.

TABLE IV
RESULTS GENERATED BY DIFFERENT METHODS ON THE MS-CMRSEG WITH THE 20% LABELED DATA. THE BEST VALUES ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Method	Metrics			
	DSC↑	Jaccard↑	HD95↓	ASD↓
SSNet (MICCAI'22) [23]	83.73±1.51	72.79±1.47	16.61±0.99	4.41±0.40
ICL (MIDL'23) [57]	84.68±1.62	74.11±1.36	13.25±0.93	1.89±0.21
BCP (CVPR'23) [38]	85.50±1.37	75.29±1.41	12.74±0.78	3.97±0.29
DCNet (MICCAI'23) [58]	84.20±1.40	73.26±1.70	9.71±0.64	1.60±0.18
DPLR (ISBI'24) [20]	84.72±1.35	74.13±1.33	7.52±0.51	1.71±0.27
AD-MT (ECCV'24) [26]	86.34±0.81	76.70±0.96	8.33±0.40	1.45±0.19
ABD (CVPR'24) [39]	87.28±0.70	77.78±1.31	8.61±0.37	4.26±0.20
SDCL (MICCAI'24) [25]	87.09±0.41	77.66±0.49	7.97±0.44	2.96±0.11
CGS (TMI'25) [22]	85.93±0.61	75.95±0.52	7.66±0.47	3.81±0.23
β-FFT (CVPR'25) [24]	87.51±0.49	78.19±0.45	6.36±0.28	1.93±0.13
UCSeg (TIP'25) [16]	87.08±0.39	77.96±0.40	5.95±0.21	1.97±0.26
HASSL (PR'26) [59]	87.14±0.37	77.06±0.41	6.35±0.30	2.12±0.24
DHPCL	**88.84±0.29**	**80.31±0.33**	**4.96±0.28**	**1.16±0.08**

TABLE V
QUANTITATIVE RESULTS GENERATED BY DIFFERENT METHODS ON THE LA DATASET WITH 10% AND 20% LABELED DATA. THE BEST VALUES ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Method	L / U	Metrics			
		DSC↑	Jaccard↑	HD95↓	ASD↓
PMT (ECCV'24) [15]	8 (10%) / 72 (90%)	89.81±1.22	81.90±1.41	7.04±2.10	2.27±0.73
Mutual (MIA'24) [12]		89.86±0.97	80.43±0.54	8.46±1.03	2.54±0.61
UCSeg (TIP'25) [16]		90.02±0.52	82.14±0.70	6.97±0.49	1.98±0.18
UnCo (TMI'25) [60]		90.13±0.67	82.54±0.47	**6.13±0.27**	2.05±0.35
HASSL (PR'26) [59]		88.94±0.52	80.01±0.64	8.97±0.21	2.56±0.16
DHPCL		**90.59±0.44**	**82.90±0.39**	6.45±0.48	**1.61±0.18**
PMT (ECCV'24) [15]	16 (20%) / 64 (80%)	90.43±0.59	82.76±0.71	6.11±0.25	1.85±0.12
Mutual (MIA'24) [12]		90.02±0.17	82.16±0.28	6.78±0.33	1.96±0.27
UCSeg (TIP'25) [16]		90.89±0.44	83.01±0.42	6.50±0.43	1.68±0.20
UnCo (TMI'25) [60]		90.79±0.35	82.94±0.36	**5.37±0.51**	1.89±0.15
HASSL (PR'26) [59]		90.58±0.39	82.87±0.40	6.12±0.57	1.90±0.14
DHPCL		**91.19±0.23**	**83.87±0.40**	5.54±0.14	**1.51±0.07**

D. Ablation Study

1) Ablation Analysis of Components: To comprehensively assess the contribution of each component in DHPCL, we conducted ablation studies on the ACDC dataset using only 10% labeled data. The evaluation begins with a semi-supervised segmentation network as the baseline and progressively incorporates the components of the proposed framework. The ablation study focuses on two major modules: (1) DGDR, which contains three key components, diffusion model, discriminator-guided rectification, and LDE; and (2) HSH-PC, which regularizes the feature space through prototype-guided contrastive learning. Performance is reported for RV, Myo, and LV.

As shown in Table VI, the baseline achieves only 86.74% average DSC with notable boundary errors. Adding the diffusion model improves pseudo-label refinement but remains limited without explicit noise guidance. The discriminator provides region-aware reliability guidance, reducing HD95 for RV and Myo to 2.31 mm and 2.72 mm, while LDE preserves fine structures and improves the average DSC to 90.28%. With HSH-PC, the full DHPCL achieves the best results, reaching 90.90% average DSC and 1.26 mm HD95. These results confirm that DGDR improves pseudo-label quality and boundary fidelity, while HSH-PC enhances semantic alignment and feature discrimination.

2) Effectiveness of the Discriminator: To verify the effectiveness of the discriminator in DGDR, we replace its confidence map with a softmax-based confidence map, where the maximum class probability is used as the pixel-wise confidence score. As shown in Table VII, the discriminator-based confidence map achieves better performance on both datasets, demonstrating that it provides more reliable pseudo-label quality estimation than simple softmax probabilities and guides diffusion rectification more effectively.

3) Effectiveness of Diffusion-Based Rectification: To verify the effectiveness of diffusion-based pseudo-label rectification, we replaced the diffusion rectifier in DGDR with three non-diffusion alternatives while keeping the backbone, HSH-PC module, and training protocol unchanged. Fully Connected CRF (FC-CRF) [61] refines pseudo-labels through dense Gaussian pairwise potentials and mean-field inference. Posterior-CRF [62] performs CRF inference in a CNN-learned feature space. DR-GAN [63] learns a noisy-to-refined pseudo-label mapping with a U-Net-like generator under adversarial supervision. Each alternative only replace the diffusion correction process, and the resulting pseudo-labels are fed into HSH-PC. The experimental results, which are shown in Table VIII, demonstrate that the proposed diffusion-based rectification achieves better segmentation accuracy and boundary metrics than all three non-diffusion alternatives.

4) Effectiveness of HSH-PC: To verify the effectiveness of HSH-PC, we replace it with a conventional Dice loss while

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

TABLE VI
QUANTITATIVE RESULTS OF ABLATION STUDY ON THE ACDC WITH THE 10% LABELED DATA. "DIFF" DENOTES THE DIFFUSION MODULE, AND "DISC" INDICATES THE DISCRIMINATOR-GUIDED RECTIFICATION. THE BEST VALUES ARE HIGHLIGHTED IN BOLD.

Components					RV				Myo				LV				Mean			
Base	Diff	Disc	LDE	HSH-PC	DSC↑	Jac↑	HD95↓	ASD↓	DSC↑	Jac↑	HD95↓	ASD↓	DSC↑	Jac↑	HD95↓	ASD↓	DSC↑	Jac↑	HD95↓	ASD↓
✓					85.13	75.25	7.58	2.02	84.25	73.12	4.96	1.21	90.83	83.77	6.44	1.80	86.74	77.38	6.33	1.67
✓	✓				88.99	81.09	3.11	0.90	85.43	74.84	6.03	1.79	90.73	83.98	8.56	2.91	88.38	79.97	5.90	1.87
✓	✓	✓			89.49	81.47	2.31	0.48	86.54	76.51	2.72	1.73	91.51	84.96	6.18	1.54	89.18	80.89	3.73	1.24
✓	✓	✓	✓		90.47	83.12	**1.30**	0.42	87.31	77.73	2.91	0.84	93.06	87.33	6.00	1.42	90.28	82.73	3.40	0.89
✓	✓	✓	✓	✓	**90.59**	**83.42**	1.37	**0.33**	**88.08**	**78.95**	**1.12**	**0.30**	**94.02**	**89.01**	**1.30**	**0.27**	**90.90**	**83.79**	**1.26**	**0.30**

TABLE VII
EFFECTIVENESS OF THE DISCRIMINATOR ON THE ACDC AND MS-CMRSEG DATASETS WITH 20% LABELED DATA.

Method	ACDC (Mean)			MS-CMRSeg		
	DSC↑	HD95↓	ASD↓	DSC↑	HD95↓	ASD↓
Softmax	88.94	2.33	1.70	86.02	7.36	2.51
Discriminator	**90.11**	**1.24**	**0.68**	**87.94**	**5.52**	**1.87**

TABLE VIII
EFFECTIVENESS OF DIFFUSION-BASED RECTIFICATION ON THE ACDC AND MS-CMRSEG DATASETS WITH 20% LABELED DATA.

Method	ACDC (Mean)			MS-CMRSeg		
	DSC↑	HD95↓	ASD↓	DSC↑	HD95↓	ASD↓
FC-CRF	86.95	6.07	3.26	85.34	7.52	3.97
Posterior-CRF	87.02	5.98	3.18	86.15	7.24	3.83
DR-GAN	88.16	3.04	2.93	86.90	6.71	3.10
Diffusion-based	**90.93**	**1.27**	**0.56**	**88.35**	**5.19**	**1.43**

keeping other components unchanged. Table IX shows that the Dice-loss variant underperforms the full model, indicating the limitation of region-level overlap supervision. In contrast, HSH-PC uses diffusion-refined prototypes and hard-soft pixel-prototype contrast to improve feature compactness and separability, providing effective discriminative supervision for semi-supervised segmentation.

5) Computational Cost and Inference Time: To evaluate the computational efficiency of DHPCL, we compare it with four representative semi-supervised segmentation methods: UniMatch [64], BCP [38], DPLR [20], and UCSeg [16]. All experiments are conducted on the ACDC dataset with 10% labeled data, with the batch size fixed to 12 for all methods. Table X reports the peak training GPU memory, total training time, and average inference time per case.

As shown in Table X, DHPCL incurs additional training overhead mainly from DGDR, leading to higher GPU memory consumption. However, these modules are used only during training and removed at inference stage. Therefore, DHPCL performs prediction with only the segmentation network, keeping its inference time comparable to existing methods while achieving stronger segmentation performance.

V. DISCUSSION

A. Analysis for the Confidence Threshold

The pseudo-label confidence threshold τ balances label reliability and sample utilization. A low threshold includes more unlabeled samples but introduces noisy pseudo-labels, degrading optimization and segmentation quality, whereas an overly high threshold improves label fidelity but reduces effective supervision and may cause underfitting. To analyze its impact, we tested τ from 0.7 to 0.95. As shown in Fig. 5, $\tau = 0.7$ introduces noisy supervision with blurred boundaries and small-structure artifacts, while $\tau > 0.9$ limits supervision diversity despite higher label reliability. Overall, the range of 0.8–0.9 achieves the best balance between pseudo-label quality and sample diversity, yielding higher values of DSC and Jaccard scores and lower values of HD95 and ASD on both ACDC and MS-CMRSeg.

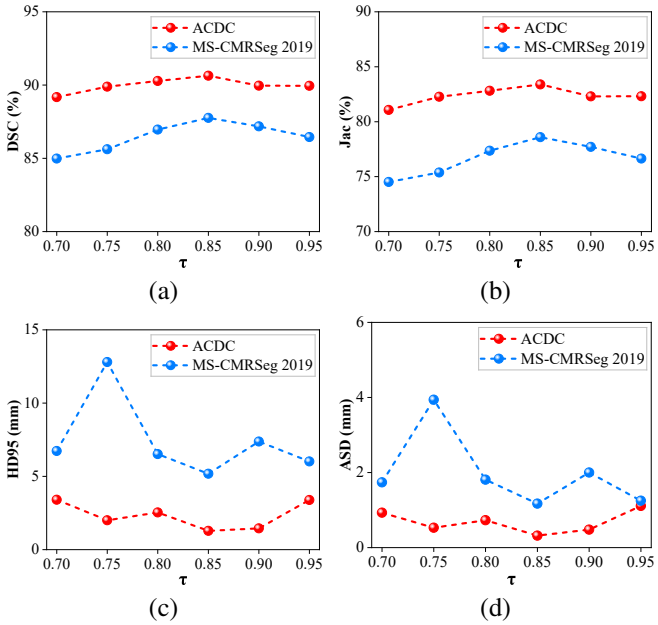


Fig. 5. Impact of the pseudo-label confidence threshold τ on segmentation performance for the ACDC and MS-CMRSeg datasets. The plots illustrate (a) DSC, (b) Jaccard, (c) HD95, and (d) ASD across different τ values.

B. Analysis for Diffusion and Sampling Steps

The diffusion and reverse sampling steps balance pseudo-label rectification quality and computational efficiency. Unlike conventional image-generation diffusion models that often require hundreds or even around one thousand time steps, DGDR only refines preliminary pseudo-labels in the latent label space, where coarse anatomical structures are already available. Thus, fewer diffusion steps are sufficient for efficient rectification. This setting is consistent with recent label-refinement studies: DiffRect [45] uses latent diffusion for progressive pseudo-label refinement, Caudo-Diff [65] calibrates pseudo-labels through

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

TABLE IX
EFFECTIVENESS OF THE PROPOSED HSH-PC MODULE ON THE ACDC DATASET WITH 10% LABELED DATA.

Method	RV				Myo				LV				Mean			
	DSC↑	Jac↑	HD95↓	ASD↓	DSC↑	Jac↑	HD95↓	ASD↓	DSC↑	Jac↑	HD95↓	ASD↓	DSC↑	Jac↑	HD95↓	ASD↓
DGDR + Dice	89.24	81.38	1.55	0.43	87.77	78.42	1.44	0.44	92.66	87.00	3.02	0.73	89.89	82.27	2.00	0.53
DGDR + HSH-PC	**90.59**	**83.42**	**1.37**	**0.33**	**88.08**	**78.95**	**1.12**	**0.30**	**94.02**	**89.01**	**1.30**	**0.27**	**90.90**	**83.79**	**1.26**	**0.30**

TABLE X
COMPARISON OF TRAINING GPU MEMORY CONSUMPTION, TOTAL TRAINING TIME, AND INFERENCE TIME.

Method	GPU Memory (GB)	Training Time (h)	Inference Time (ms)
UniMatch [64]	4.95	2.47	29.2
BCP [38]	1.30	2.07	72.2
DPLR [20]	4.65	3.04	65.0
UCSeg [16]	5.62	6.80	106.1
DHPCL	8.80	4.59	59.1

deterministic latent diffusion, and DLD [66] shows that label-level diffusion can be accelerated with only a few inference steps. Based on these observations, we restrict the diffusion steps to [10, 35] and set the reverse sampling steps proportionally. To empirically determine the optimal configuration, we evaluate the impact of varying these steps on segmentation performance, with the results presented in Fig. 6.

In Fig. 6, too few diffusion and sampling steps leave residual pseudo-label noise, while moderate steps improve boundary precision and structural consistency. Excessive steps may perturb reliable pseudo-labels and introduce gradient noise, causing slight performance degradation, particularly on the MS-CMRSeg. To balance the accuracy, efficiency, and stability of DHPCL, we set the diffusion and reverse sampling steps to 20 and 4, respectively.

C. Clinical Impact

The clinical relevance of DHPCL stems from its joint enhancement of pseudo-label reliability and feature discriminability. DGDR adaptively corrects noisy pseudo-labels and structurally inconsistent regions through discriminator-guided diffusion rectification, while HSH-PC aligns pixel features with refined class prototypes to improve boundary delineation and semantic consistency. These improvements can provide more reliable anatomical contours for downstream quantitative analysis, such as cardiac structural measurement and morphological assessment, thereby reducing uncertainty in segmentation-based clinical interpretation. Moreover, DHPCL effectively leverages unlabeled medical images and retains only the segmentation network during inference, reducing annotation dependence without adding testing burden.

D. Failure Case

Fig. 7 presents representative failure cases of DHPCL, where errors mainly occur in relatively small LV regions, as indicated by the red arrows. When the LV occupies only a small portion of the image, its boundary is more susceptible to weak contrast and surrounding tissues, leading to local over-segmentation, under-segmentation, or slight boundary deviation. This suggests that DGDR may be limited when initial

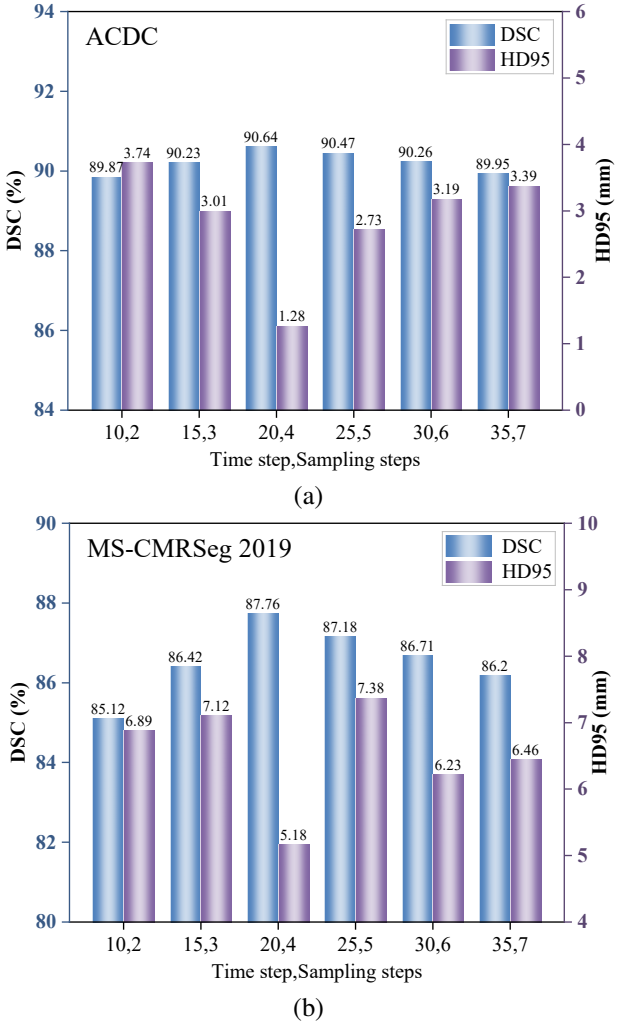


Fig. 6. The figure summarizes the DSC and HD95 obtained under different diffusion-sampling step combinations. (a) Performance on the ACDC. (b) Performance on the MS-CMRSeg.

pseudo-labels lack sufficient structural cues, while HSH-PC may also suffer from less representative prototypes due to the limited number of reliable LV pixels. Therefore, small LV regions remain a challenging scenario for the DHPCL.

Fundamentally, this reveals a limitation of DHPCL: current prototype representations may occasionally struggle to capture complex and small target distributions in extreme edge cases. In future work, we intend to introduce uncertainty-aware dynamic multi-pole prototypes to model the intra-class variations of small anatomical structures, aiming to mitigate the representational limitations caused by sparse reliable pixels.

VI. CONCLUSION

We propose a novel framework, DHPCL, for semi-supervised medical image segmentation, and it intrinsically

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

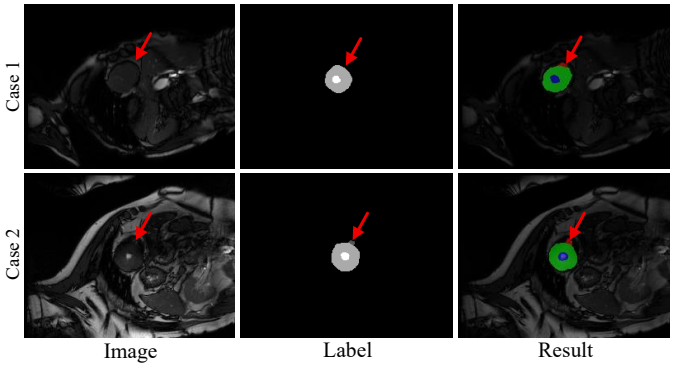


Fig. 7. Representative failure cases of DHPCL in small-target LV segmentation. Red arrows indicate relatively small LV regions where segmentation errors occur.

links pseudo-label refinement with feature representation learning. Instead of treating label quality and feature learning in isolation, our approach establishes a mutually reinforcing mechanism where enhanced pseudo-label reliability directly contributes to stronger semantic feature consistency. Specifically, DGDR performs fine-grained optimization of noisy pseudo-labels through iterative denoising and reconstruction, leading to markedly improved boundary smoothness and label coherence. Meanwhile, HSH-PC constructs stable class-wise prototypes from refined pseudo-labels and enforces hard-soft pixel-prototype alignment, reducing feature ambiguity and enhancing semantic discrimination in uncertain regions. Extensive experiments across three diverse datasets demonstrate DHPCL superior performance under limited annotations.

REFERENCES

[1] S. Iqbal, T. M. Khan, S. S. Naqvi, A. Naveed, and E. Meijering, "Tbconvl-net: A hybrid deep learning architecture for robust medical image segmentation," *Pattern Recognition*, vol. 158, p. 111028, 2025.

[2] M. M. Rahman, M. Munir, and R. Marculescu, "Emcad: Efficient multi-scale convolutional attention decoding for medical image segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 11 769–11 779.

[3] Y. Du, F. Bai, T. Huang, and B. Zhao, "Segvol: Universal and interactive volumetric medical image segmentation," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 110 746–110 783.

[4] R. Azad, E. K. Aghdam, A. Rauland, Y. Jia, A. H. Avval, A. Bozorgpour, S. Karimijafarbigloo, J. P. Cohen, E. Adeli, and D. Merhof, "Medical image segmentation review: The success of u-net," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 12, pp. 10 076–10 095, 2024.

[5] M. M. Rahman and R. Marculescu, "Mk-unet: Multi-kernel lightweight cnn for medical image segmentation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 1042–1051.

[6] X. Guo, X. Lin, X. Yang, L. Yu, K.-T. Cheng, and Z. Yan, "Uctnet: Uncertainty-guided cnn-transformer hybrid networks for medical image segmentation," *Pattern Recognition*, vol. 152, p. 110491, 2024.

[7] Z. Li, Y. Zheng, D. Shan, S. Yang, Q. Li, B. Wang, Y. Zhang, Q. Hong, and D. Shen, "Scribformer: Transformer makes cnn work better for scribble-based medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 43, no. 6, pp. 2254–2265, 2024.

[8] S. Perera, P. Navard, and A. Yilmaz, "Segformer3d: an efficient transformer for 3d medical image segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 4981–4988.

[9] S. Ren and X. Li, "Hresformer: Hybrid residual transformer for volumetric medical image segmentation," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 6, pp. 10 558–10 566, 2025.

[10] B. Zhang, Z. Zheng, Y. Zhao, Y. Shen, and M. Sun, "Mcbtnet: Multi-feature fusion cnn and bi-level routing attention transformer-based medical image segmentation network," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 7, pp. 5069–5082, 2025.

[11] H. Huang, Y. Huang, S. Xie, L. Lin, R. Tong, Y.-W. Chen, Y. Li, and Y. Zheng, "Combinatorial cnn-transformer learning with manifold constraints for semi-supervised medical image segmentation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024, pp. 2330–2338.

[12] J. Su, Z. Luo, S. Lian, D. Lin, and S. Li, "Mutual learning with reliable pseudo label for semi-supervised medical image segmentation," *Medical Image Analysis*, vol. 94, p. 103111, 2024.

[13] Q. Pan, W. Qiao, J. Lou, B. Ji, and S. Li, "Dusss: Dual semantic similarity-supervised vision-language model for semi-supervised medical image segmentation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, pp. 6299–6307.

[14] Q. Zeng, Z. Lu, Y. Xie, and Y. Xia, "Pick: Predict and mask for semi-supervised medical image segmentation," *International Journal of Computer Vision*, vol. 133, no. 6, p. 3296–3311, 2025.

[15] N. Gao, S. Zhou, L. Wang, and N. Zheng, "Pmt: Progressive mean teacher via exploring temporal consistency for semi-supervised medical image segmentation," in *European Conference on Computer Vision*. Springer, 2024, pp. 144–160.

[16] K. Huang, T. Zhou, H. Fu, Y. Zhang, Y. Zhou, and X.-J. Wu, "Uncertainty-aware cross-training for semi-supervised medical image segmentation," *IEEE Transactions on Image Processing*, vol. 34, pp. 5543–5556, 2025.

[17] H. Liu, P. Ren, Y. Yuan, C. Song, and F. Luo, "Uncertainty global contrastive learning framework for semi-supervised medical image segmentation," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 1, pp. 433–442, 2025.

[18] H. Mai, R. Sun, T. Zhang, and F. Wu, "Rankmatch: Exploring the better consistency regularization for semi-supervised semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 3391–3401.

[19] M. N. Rizve, K. Duarte, Y. S. Rawat, and M. Shah, "In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning," in *International Conference on Learning Representations*, 2021, pp. 7533–7553.

[20] Y. Lu, Z. Zhang, Y. Zhang, Y. Ou, and X. Sun, "Semi-supervised medical image segmentation via dynamic pseudo-label refinement," in *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*. IEEE, 2024, pp. 1–5.

[21] Q. Ma, J. Zhang, Z. Li, L. Qi, Q. Yu, and Y. Shi, "Steady progress beats stagnation: Mutual aid of foundation and conventional models in mixed domain semi-supervised medical image segmentation," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 5175–5185.

[22] Y. Wang, Z. Li, L. Qi, Q. Yu, Y. Shi, and Y. Gao, "Balancing multi-target semi-supervised medical image segmentation with collaborative generalist and specialists," *IEEE Transactions on Medical Imaging*, vol. 44, no. 7, pp. 3025–3037, 2025.

[23] Y. Wu, Z. Wu, Q. Wu, Z. Ge, and J. Cai, "Exploring smoothness and class-separation for semi-supervised medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2022, pp. 34–43.

[24] M. Hu, J. Yin, Z. Ma, J. Ma, F. Zhu, B. Wu, Y. Wen, M. Wu, C. Hu, B. Hu *et al.*, "beta-fft: Nonlinear interpolation and differentiated training strategies for semi-supervised medical image segmentation," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 30 839–30 849.

[25] B. Song and Q. Wang, "Sdcl: Students discrepancy-informed correction learning for semi-supervised medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2024, pp. 567–577.

[26] Z. Zhao, Z. Wang, L. Wang, D. Yu, Y. Yuan, and L. Zhou, "Alternate diverse teaching for semi-supervised medical image segmentation," in *European Conference on Computer Vision*. Springer, 2024, pp. 227–243.

[27] Z. Zhang, H. Zhou, X. Shi, R. Ran, C. Tian, and F. Zhou, "An evidential-enhanced tri-branch consistency learning method for semi-supervised medical image segmentation," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–13, 2024.

[28] Q. Zeng, Y. Xie, Z. Lu, M. Lu, J. Zhang, and Y. Xia, "Consistency-guided differential decoding for enhancing semi-supervised medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 44, no. 1, pp. 44–56, 2024.

[29] H. Wu, X. Li, and K.-T. Cheng, "Exploring feature representation learning for semi-supervised medical image segmentation," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 11, pp. 16 589–16 601, 2023.

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2026.3734933

[30] G. Liang, Q. Zhou, J. Chen, Z. Wang, and C. Yao, "Advancing medical image segmentation via self-supervised instance-adaptive prototype learning," in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*. International Joint Conferences on Artificial Intelligence Organization, 2024, pp. 1056–1064.

[31] Y. Wang, B. Xiao, X. Bi, W. Li, and X. Gao, "Boundary-aware prototype in semi-supervised medical image segmentation," *IEEE Transactions on Image Processing*, vol. 33, pp. 5456–5467, 2024.

[32] Z. Lv, Z. Wu, and J. Zhu, "Clustering-guided contrastive prototype learning: Towards semi-supervised medical image segmentation," *Pattern Recognition*, vol. 171, p. 112321, 2026.

[33] D.-H. Lee *et al.*, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *Workshop on challenges in representation learning, ICML*, vol. 3 no. 2. Atlanta, 2013, p. 896.

[34] H. Basak and Z. Yin, "Pseudo-label guided contrastive learning for semi-supervised medical image segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 19 786–19 797.

[35] J. Miao, S. P. Zhou, G. Q. Zhou, K. N. Wang, M. Yang, S. Zhou, and Y. Chen, "Sc-ssl: Self-correcting collaborative and contrastive co-training model for semi-supervised medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 43, no. 4, pp. 1347–1364, 2023.

[36] Y. Yang, J. Zhuang, G. Sun, R. Wang, and J. Su, "Boundary-guided contrastive learning for semi-supervised medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 44, no. 7, pp. 2973–2988, 2025.

[37] K. Sohn, D. Berthelot, N. Carlini, Z. Zhang, H. Zhang, C. A. Raffel, E. D. Cubuk, A. Kurakin, and C. L. Li, "Fixmatch: Simplifying semi-supervised learning with consistency and confidence," *Advances in Neural Information Processing Systems*, vol. 33, pp. 596–608, 2020.

[38] Y. Bai, D. Chen, Q. Li, W. Shen, and Y. Wang, "Bidirectional copy-paste for semi-supervised medical image segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 11 514–11 524.

[39] H. Chi, J. Pang, B. Zhang, and W. Liu, "Adaptive bidirectional displacement for semi-supervised medical image segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 4070–4080.

[40] W. Huang, L. Zhang, Z. Wang, and Y. Wang, "Gapmatch: Bridging instance and model perturbations for enhanced semi-supervised medical image segmentation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, pp. 17 458–17 466.

[41] Y. Yang, G. Sun, T. Zhang, R. Wang, and J. Su, "Semi-supervised medical image segmentation via weak-to-strong perturbation consistency and edge-aware contrastive representation," *Medical Image Analysis*, vol. 101, p. 103450, 2025.

[42] J. Wu, R. Fu, H. Fang, Y. Zhang, Y. Yang, H. Xiong, H. Liu, and Y. Xu, "Medsegdiff: Medical image segmentation with diffusion probabilistic model," in *Medical Imaging with Deep Learning*. PMLR, 2024, pp. 1623–1639.

[43] J. Wu, W. Ji, H. Fu, M. Xu, Y. Jin, and Y. Xu, "Medsegdiff-v2: Diffusion-based medical image segmentation with transformer," in *Proceedings of the AAAI conference on artificial intelligence*, 2024, pp. 6030–6038.

[44] Y. Yao, X. Duan, A. Qu, M. Chen, J. Chen, and L. Chen, "Dfcg: A dual-frequency cascade graph model for semi-supervised ultrasound image segmentation with diffusion model," *Knowledge-Based Systems*, vol. 300, p. 112261, 2024.

[45] X. Liu, W. Li, and Y. Yuan, "Diffrect: Latent diffusion label rectification for semi-supervised medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2024, pp. 56–66.

[46] Y. Chen, Y. Liu, M. Lu, L. Fu, and F. Yang, "Multi-consistency for semi-supervised medical image segmentation via diffusion models," *Pattern Recognition*, vol. 161, p. 111216, 2025.

[47] L. Zhang, F. Wu, K. Bronik, and B. W. Papiez, "Diffuseg: domain-driven diffusion for medical image segmentation," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 5, pp. 3619–3631, 2025.

[48] R. Wang, Q. Zhou, and G. Zheng, "Few-shot medical image segmentation regularized with self-reference and contrastive learning," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2022, pp. 514–523.

[49] Y. Song, X. Du, Y. Zhang, and C. Xu, "Multi-shot prototype contrastive learning and semantic reasoning for medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 578–588.

[50] F. Sun, Z. Luo, and S. Li, "Boundary difference over union loss for medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 292–301.

[51] O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, X. Yang, P.-A. Heng, I. Cetin, K. Lekadir, O. Camara, M. A. Gonzalez Ballester, G. Sanroma, S. Napel, S. Petersen, G. Tziritas, E. Grinias, M. Khened, V. A. Kollerathu, G. Krishnamurthi, M.-M. Rohé, X. Pennec, M. Sermesant, F. Isensee, P. Jäger, K. H. Maier-Hein, P. M. Full, I. Wolf, S. Engelhardt, C. F. Baumgartner, L. M. Koch, J. M. Wolterink, I. Išgum, Y. Jang, Y. Hong, J. Patravali, S. Jain, O. Humbert, and P.-M. Jodoin, "Deep learning techniques for automatic mri cardiac multi-structures segmentation and diagnosis: Is the problem solved?" *IEEE Transactions on Medical Imaging*, vol. 37, no. 11, pp. 2514–2525, 2018.

[52] X. Zhuang, "Multivariate mixture model for myocardial segmentation combining multi-source images," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 12, pp. 2933–2946, 2019.

[53] F. Wu and X. Zhuang, "Minimizing estimated risks on unlabeled data: A new formulation for semi-supervised medical image segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 5, pp. 6021–6036, 2023.

[54] Z. Xiong, Q. Xia, Z. Hu, N. Huang, C. Bian, Y. Zheng, S. Vesal, N. Ravikumar, A. Maier, X. Yang, P.-A. Heng, D. Ni, C. Li, Q. Tong, W. Si, E. Puybareau, Y. Khoudli, T. Géraud, C. Chen, W. Bai, D. Rueckert, L. Xu, X. Zhuang, X. Luo, S. Jia, M. Sermesant, Y. Liu, K. Wang, D. Borra, A. Masci, C. Corsi, C. de Vente, M. Veta, R. Karim, C. J. Preetha, S. Engelhardt, M. Qiao, Y. Wang, Q. Tao, M. Nuñez-Garcia, O. Camara, N. Savioli, P. Lamata, and J. Zhao, "A global benchmark of algorithms for segmenting the left atrium from late gadolinium-enhanced cardiac magnetic resonance imaging," *Medical Image Analysis*, vol. 67, p. 101832, 2021.

[55] X. Luo, M. Hu, T. Song, G. Wang, and S. Zhang, "Semi-supervised medical image segmentation via cross teaching between cnn and transformer," in *International Conference on Medical Imaging with Deep Learning*. PMLR, 2022, pp. 820–833.

[56] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature methods*, vol. 18, no. 2, pp. 203–211, 2021.

[57] Y. Zhu, J. Yang, S. Liu, and R. Zhang, "Inherent consistent learning for accurate semi-supervised medical image segmentation," in *Medical Imaging with Deep Learning*, vol. 227. PMLR, 2023, pp. 1581–1601.

[58] F. Chen, J. Fei, Y. Chen, and C. Huang, "Decoupled consistency for semi-supervised medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 551–561.

[59] B. Zheng, W. Zhao, W. Liu, Z. He, C. Qin, and H. Yang, "Semi-supervised medical image segmentation via pseudo-labeling refinement and dual-adaptive adjustment schemes," *Pattern Recognition*, vol. 171, p. 112310, 2026.

[60] X. Zeng, S. Xiong, J. Xu, G. Du, and Y. Rong, "Uncertainty co-estimator for improving semi-supervised medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 44, no. 9, pp. 3870–3881, 2025.

[61] P. Krähenbühl and V. Koltun, "Efficient inference in fully connected crfs with gaussian edge potentials," in *Advances in Neural Information Processing Systems*, vol. 24, 2011, pp. 109–117.

[62] S. Chen, Z. Sedghi Gamechi, F. Dubost, G. van Tulder, and M. de Bruijne, "An end-to-end approach to segmentation in medical images with cnn and posterior-crf," *Medical Image Analysis*, vol. 76, p. 102311, 2022.

[63] K. Liao, C. Lin, Y. Zhao, and M. Gabbouj, "Dr-gan: Automatic radial distortion rectification using conditional gan in real-time," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 30, no. 3, pp. 725–733, 2020.

[64] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 7236–7246.

[65] B. Yu and Y. Liu, "Caudo-diff: Diffusion calibrated pseudo labels in guided latent space for minimally supervised medical segmentation," *Pattern Recognition*, vol. 170, p. 112007, 2026.

[66] S. Hou, G. Jiang, J. Zhang, S. Yang, H. Guo, Y. Guo, and W. Wang, "Directional label diffusion model for learning from noisy labels," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 25 738–25 748.

Authorized licensed use limited to: National Huaqiao University. Downloaded on October 03,2026 at 09:50:11 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.